DSA 8020 Project 1

saransh rakshak

Contents


```
library(dplyr)
```

```
##
## Attaching package: 'dplyr'
```

```
## The following objects are masked from 'package:stats':
##
##     filter, lag
```

```
## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(ggplot2)
library(MASS)
```

```
##
## Attaching package: 'MASS'
```

```
## The following object is masked from 'package:dplyr':
##
##     select
```

```r
library(mgcv)
```

```
## Loading required package: nlme
```

```
##
## Attaching package: 'nlme'
```

```
## The following object is masked from 'package:dplyr':
##
##     collapse
```

```
## This is mgcv 1.9-1. For overview type 'help("mgcv-package")'.
```

```r
library(car)
```

```
## Loading required package: carData
```

```
##
## Attaching package: 'car'
```

```
## The following object is masked from 'package:dplyr':
##
##     recode
```

Instructions

For each problem, your answer should include the following elements:

- A descriptive summary of the data using summary statistics and at least one plot/graph or table. Give your description in paragraph form.

- An appropriate inferential analysis to address the question. This may be a confidence interval, hypothesis test, or both. You can use the α value/confidence level of your choice. If you perform a hypothesis test, state the hypotheses, test statistic, p-value, decision, and conclusion. If you create a confidence interval, state the confidence level and provide an interpretation of the interval.
- A thoughtful conclusion that summarizes your findings. You should also discuss any limitations or complications of your analysis. This should include the following:
  - An assessment of whether the modeling assumptions are reasonable.
  - Discussion of any unusual features of the data. For example, do outliers/high leverage points affect your results?

Problem 1: Gala

- Perform a detailed regression analysis using the gala data set. Use Endemics as the response variable, excluding Species from this analysis, to address the following questions:

Code:

```
library(faraway)
```

```
##
## Attaching package: 'faraway'

## The following objects are masked from 'package:car':
##
##     logit, vif
```

```
data(gala)

# Excluding Species from analysis
gala <- subset(gala, select = -Species)
head(gala)
```

```
##              Endemics  Area Elevation Nearest Scruz Adjacent
## Baltra             23 25.09       346     0.6   0.6     1.84
## Bartolome          21  1.24       109     0.6  26.3   572.33
## Caldwell            3  0.21       114     2.8  58.7     0.78
## Champion            9  0.10        46     1.9  47.4     0.18
## Coamano             1  0.05        77     1.9   1.9   903.82
## Daphne.Major       11  0.34       119     8.0   8.0     1.84
```

Data Format

Source: M. P. Johnson and P. H. Raven (1973) "Species number and endemism: The Galapagos Archipelago revisited" Science, 179, 893-895

The dataset contains the following variables

Species (Excluded from analysis) - the number of plant species found on the island

Endemics - the number of endemic species

Area - the area of the island (km^2)

Elevation - the highest elevation of the island (m)

Nearest - the distance from the nearest island (km)

Scruz - the distance from Santa Cruz island (km)

Adjacent - the area of the adjacent island (square km)

---

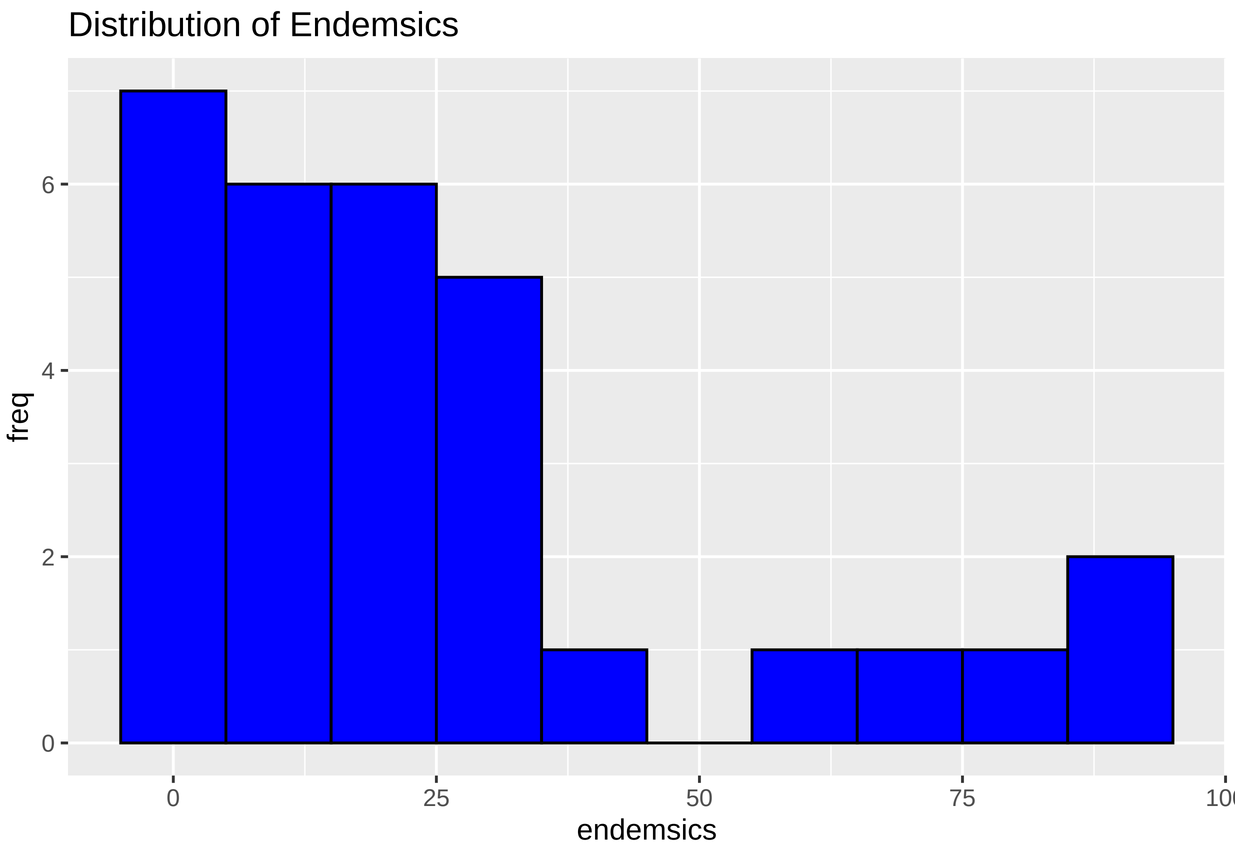
1. Provide a descriptive summary of the data set.

Code:

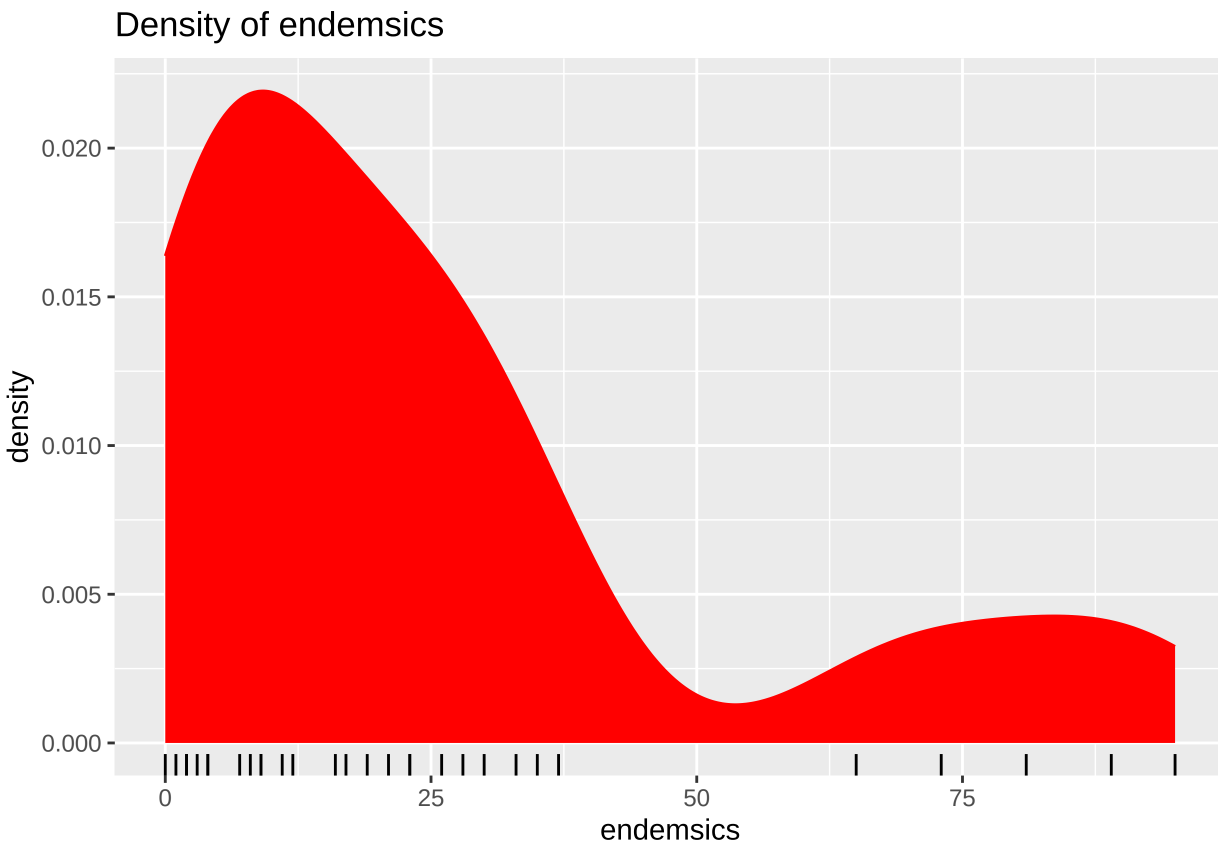
```
summary(gala)
```

```
##     Endemics          Area             Elevation          Nearest
##  Min.   : 0.00   Min.   :   0.010   Min.   :  25.00   Min.   : 0.20
##  1st Qu.: 7.25   1st Qu.:   0.258   1st Qu.:  97.75   1st Qu.: 0.80
##  Median :18.00   Median :   2.590   Median : 192.00   Median : 3.05
##  Mean   :26.10   Mean   : 261.709   Mean   : 368.03   Mean   :10.06
##  3rd Qu.:32.25   3rd Qu.:  59.237   3rd Qu.: 435.25   3rd Qu.:10.03
##  Max.   :95.00   Max.   :4669.320   Max.   :1707.00   Max.   :47.40
##      Scruz            Adjacent
##  Min.   :  0.00   Min.   :   0.03
##  1st Qu.: 11.03   1st Qu.:   0.52
##  Median : 46.65   Median :   2.59
##  Mean   : 56.98   Mean   : 261.10
##  3rd Qu.: 81.08   3rd Qu.:  59.24
##  Max.   :290.20   Max.   :4669.32
```

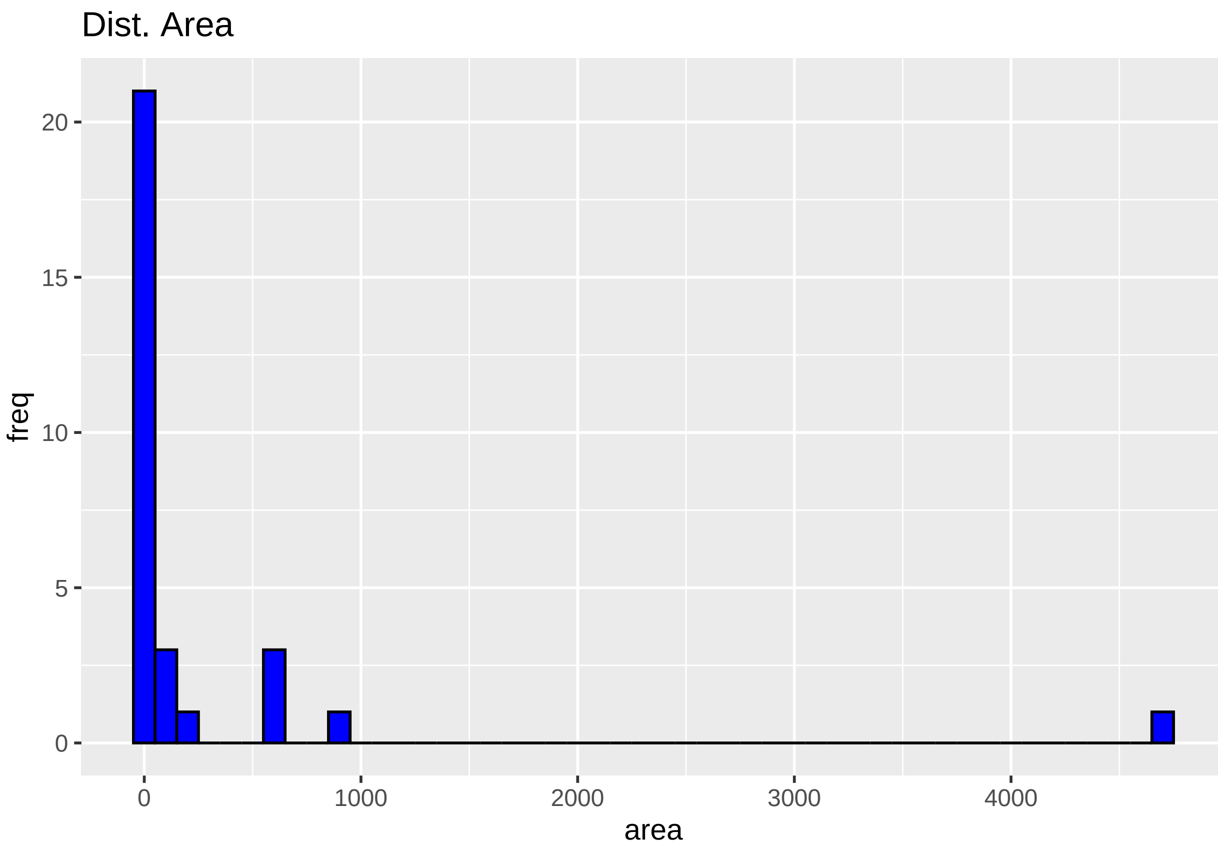
```
# plot for endemics (response var)
# hist
ggplot(gala, aes(x = Endemics)) +
  geom_histogram(binwidth = 10, color = "black", fill = "blue") +
  labs(title = "Distribution of Endemsics", x = "endemsics", y = "freq")
```



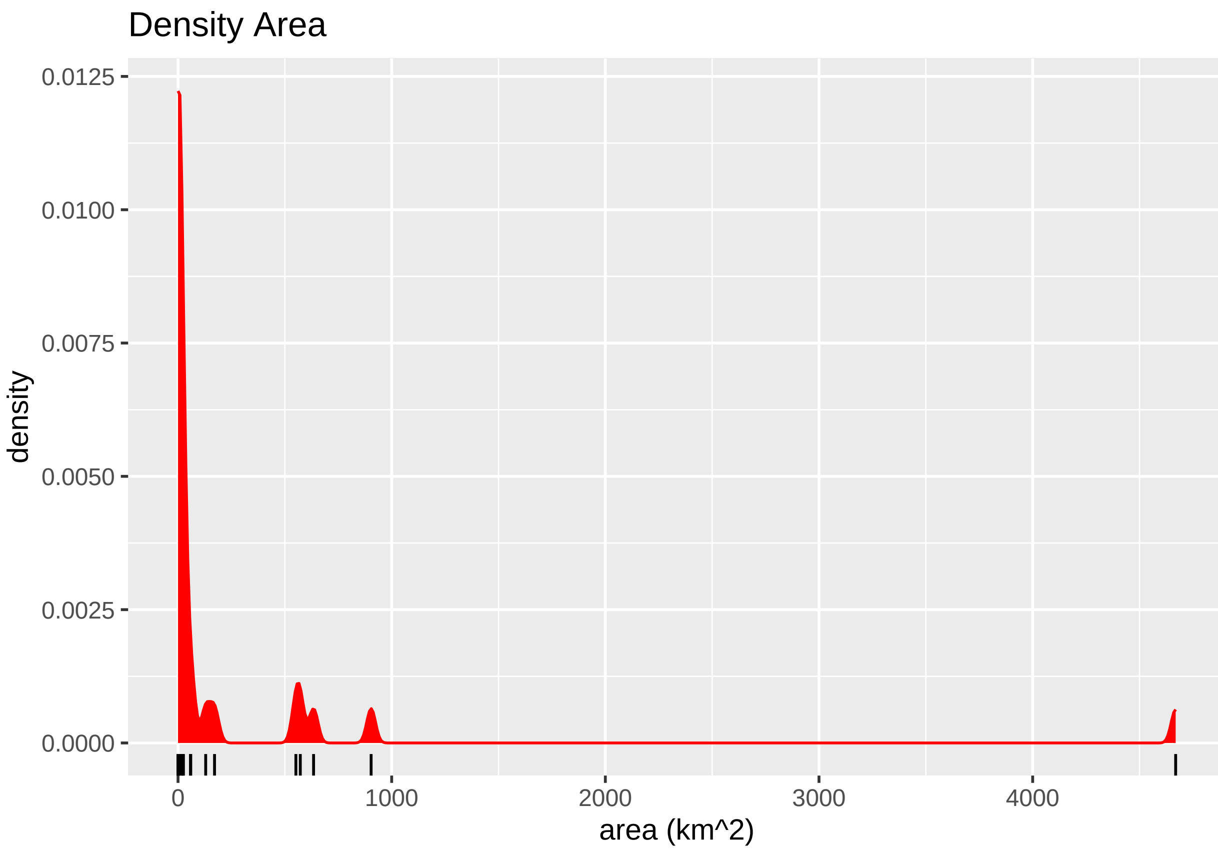

```r
# density
ggplot(gala, aes(x = Endemics)) +
  geom_density(color = "red", fill = "red") +
  geom_rug() +
  labs(title = "Density of endemsics", x = "endemsics")
```



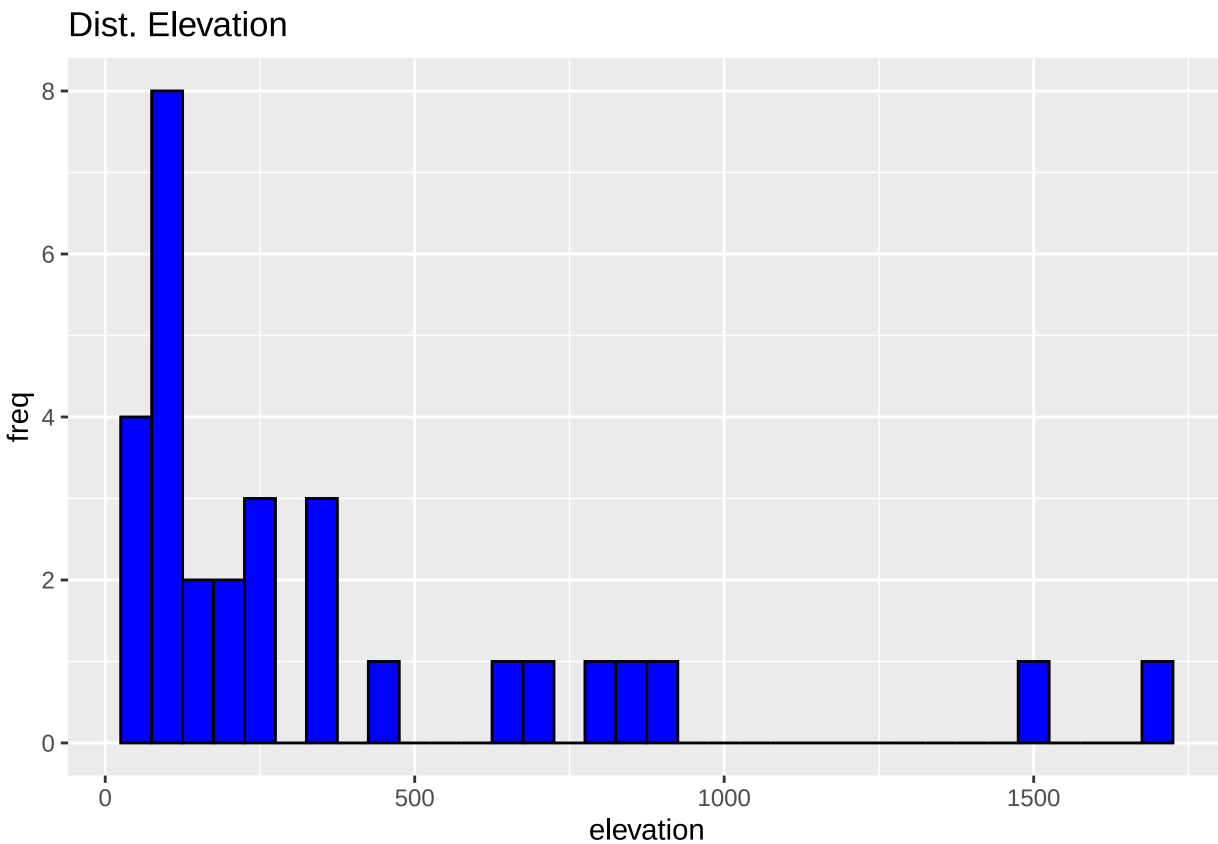

```r
# plotting dist and density for Area
ggplot(gala, aes(x = Area)) +
  geom_histogram(binwidth = 100, color = "black", fill = "blue") +
  labs(title = "Dist. Area", x = "area", y = "freq")
```



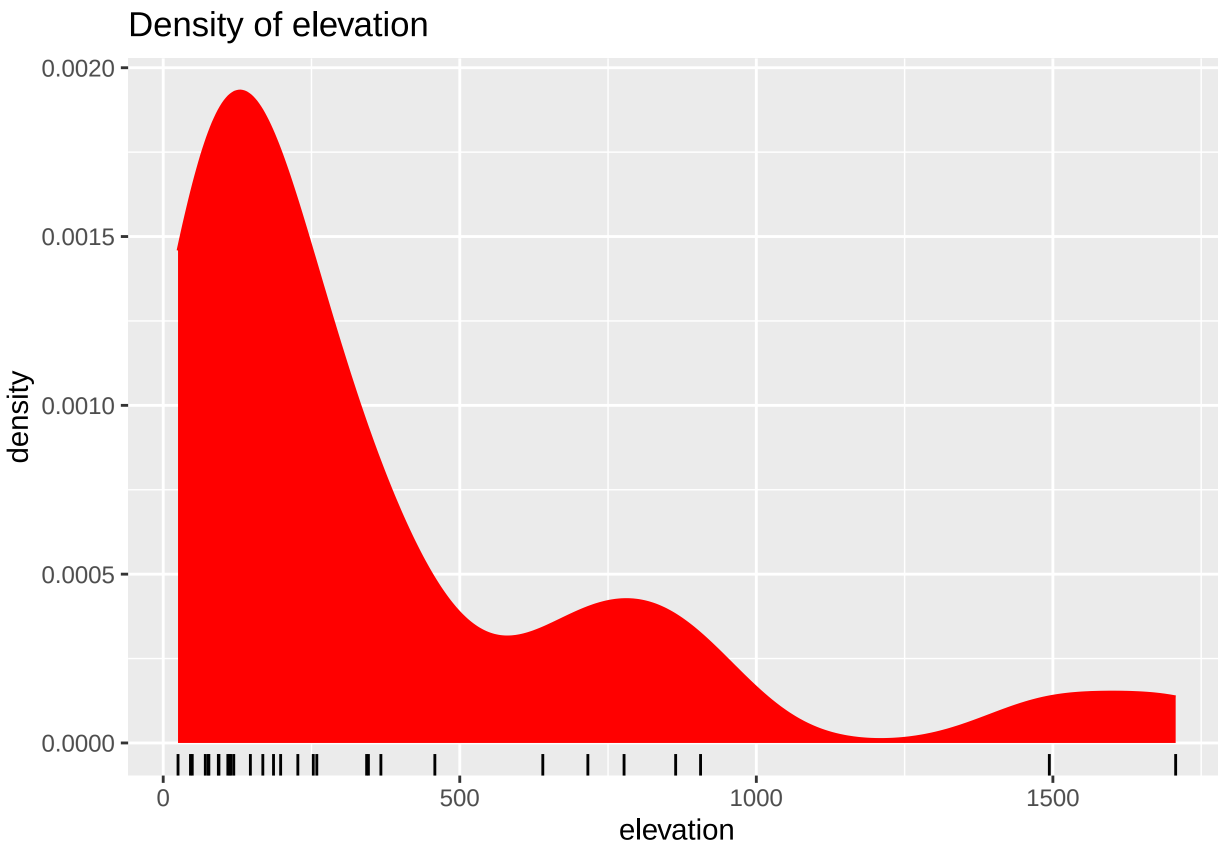

```r
ggplot(gala, aes(x = Area)) +
  geom_density(color = "red", fill = "red") +
  geom_rug() +
  labs(title = "Density Area", x = "area (km^2)")
```



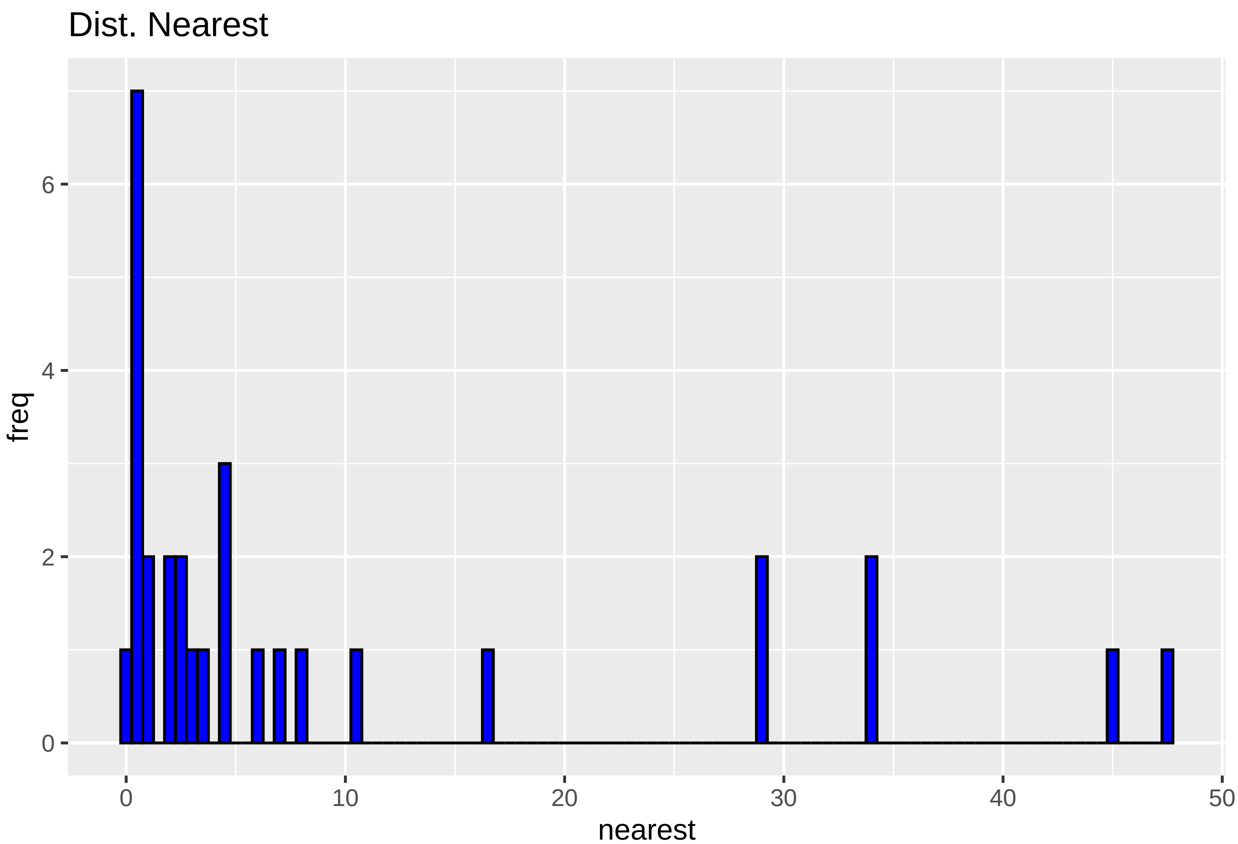

```r
# plots for Elevation
ggplot(gala, aes(x = Elevation)) +
  geom_histogram(binwidth = 50, color = "black", fill = "blue") +
  labs(title = "Dist. Elevation", x = "elevation", y = "freq")
```



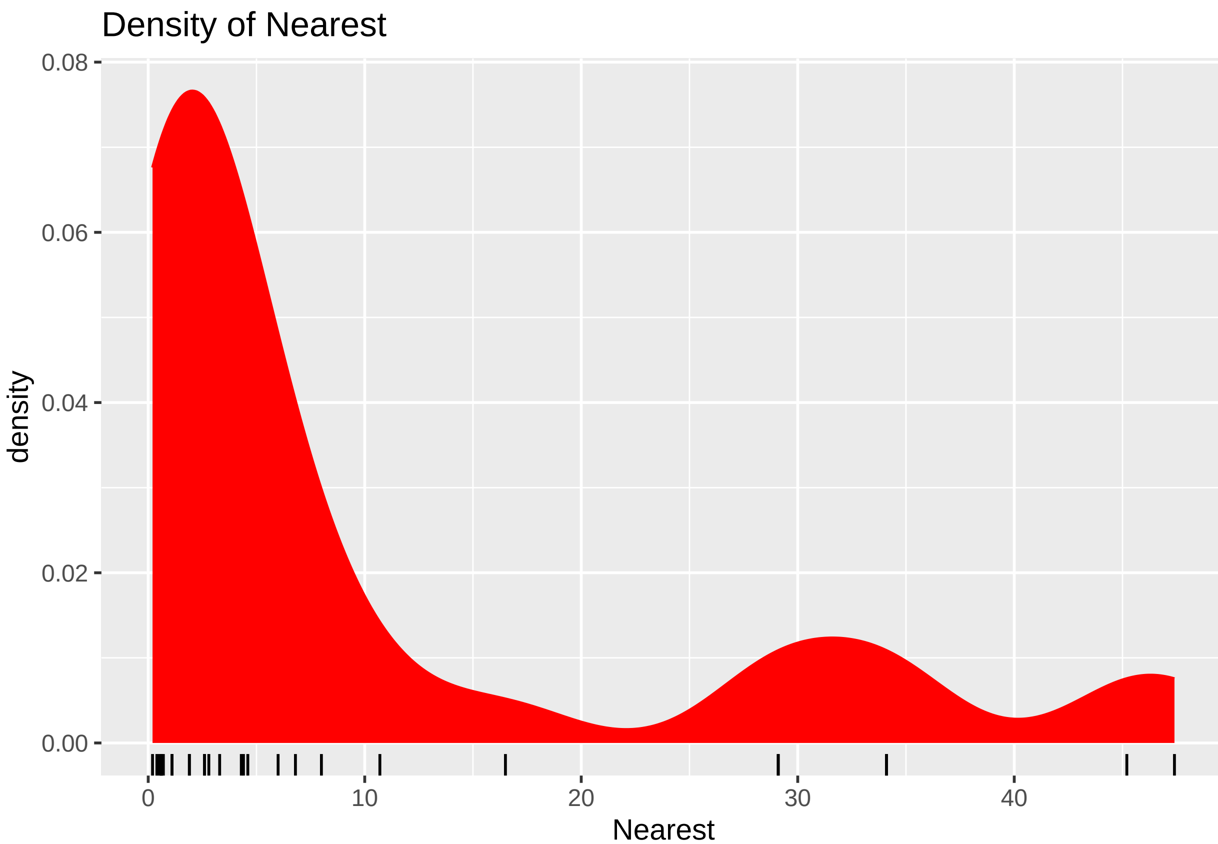

```r
ggplot(gala, aes(x = Elevation)) +
  geom_density(color = "red", fill = "red") + geom_rug() +
  labs(title = "Density of elevation", x = "elevation")
```



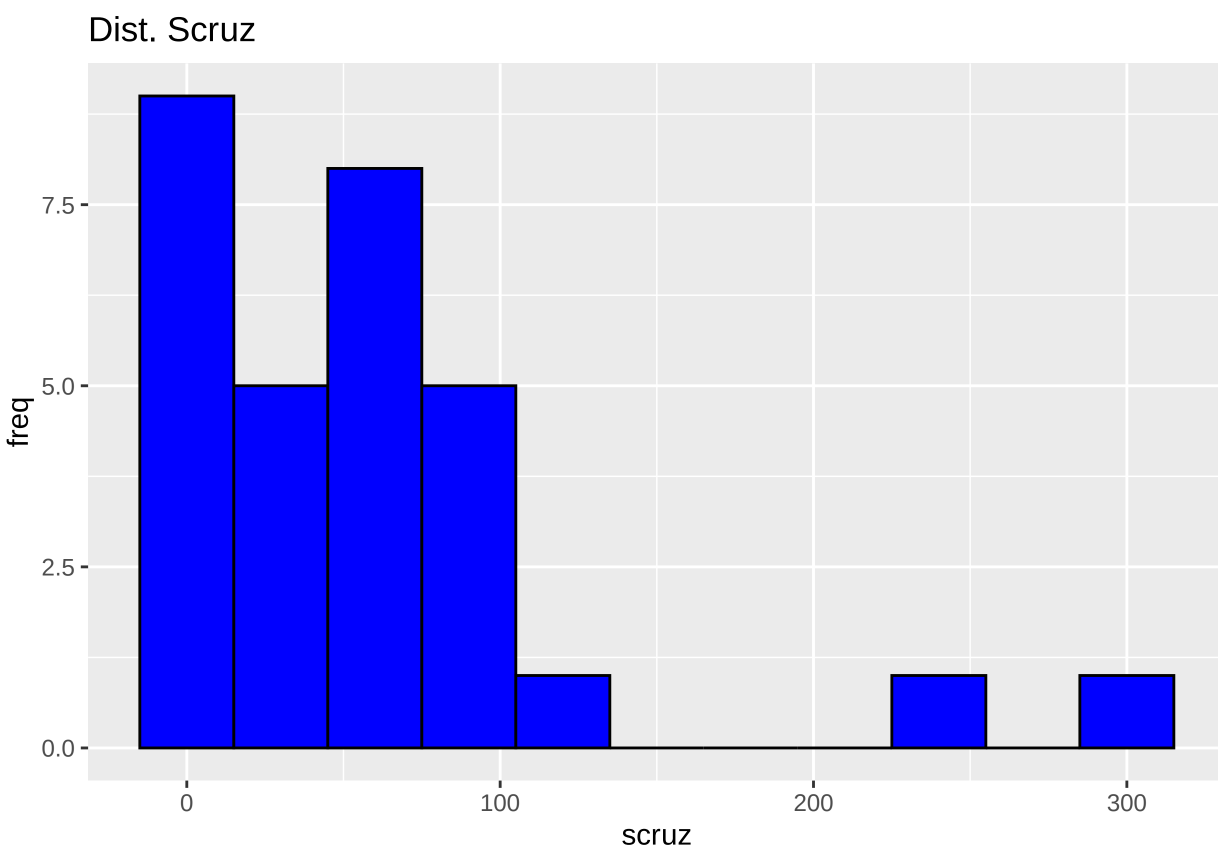

```r
# Nearest
ggplot(gala, aes(x = Nearest)) +
  geom_histogram(binwidth = 0.5, color = "black", fill = "blue") +
  labs(title = "Dist. Nearest", x = "nearest", y = "freq")
```



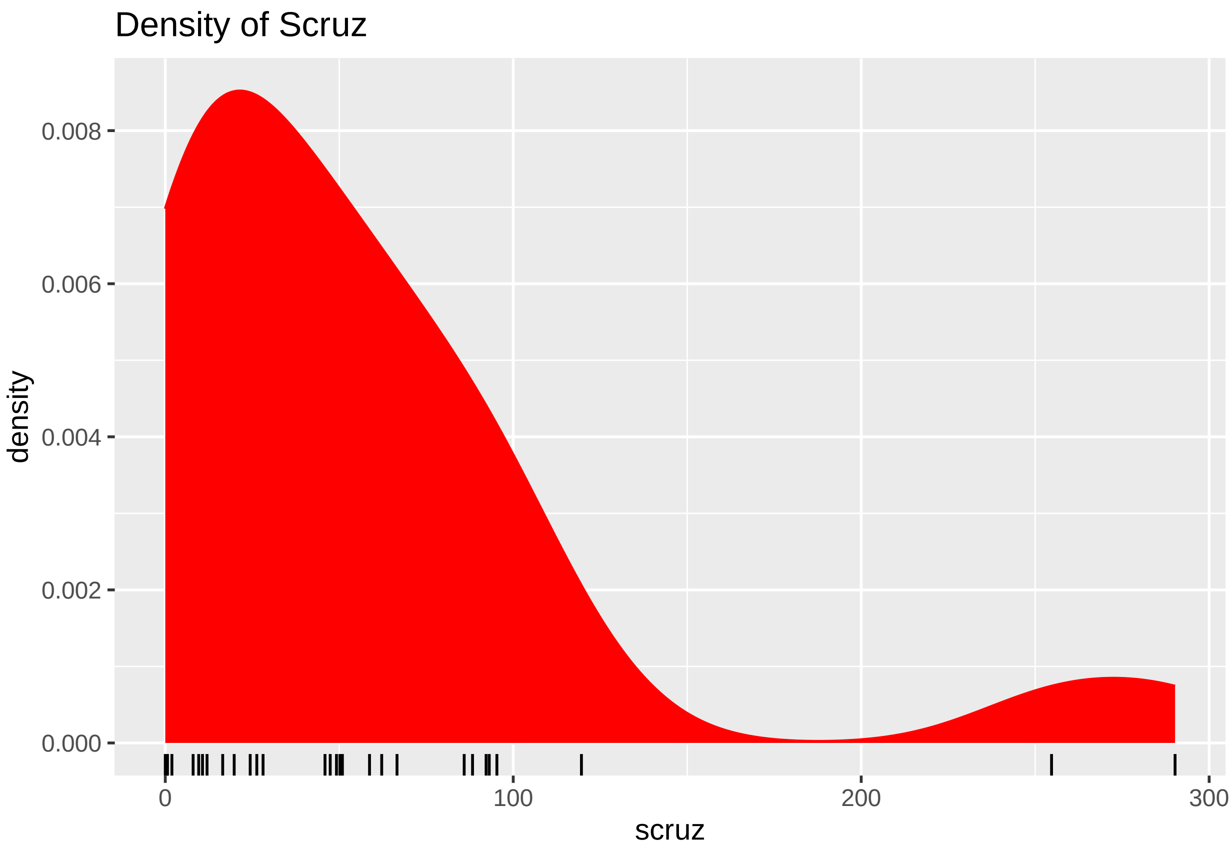

```r
ggplot(gala, aes(x = Nearest)) +
  geom_density(color = "red", fill = "red") +
  geom_rug() +
  labs(title = "Density of Nearest", x = "Nearest")
```



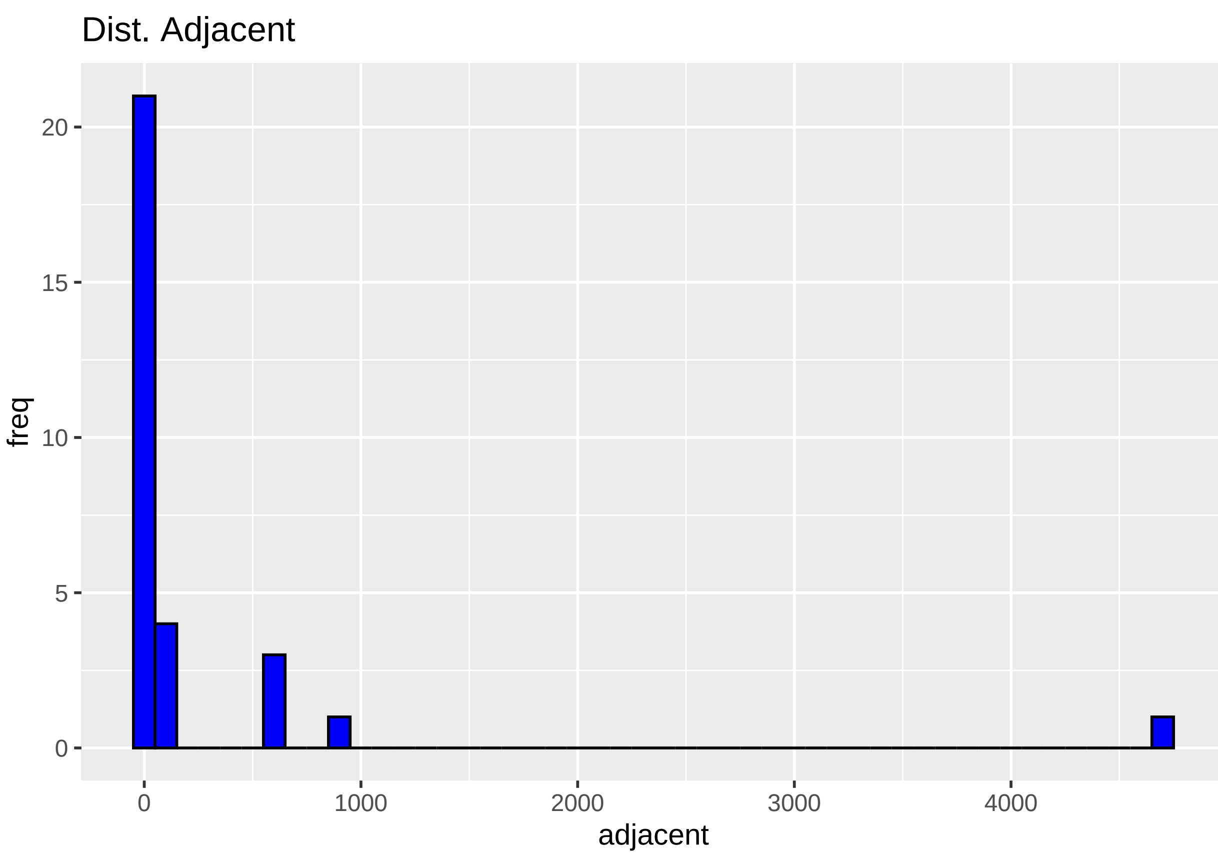

```r
# Scruz
ggplot(gala, aes(x = Scruz)) +
  geom_histogram(binwidth = 30, color = "black", fill = "blue") +
  labs(title = "Dist. Scruz", x = "scruz", y = "freq")
```



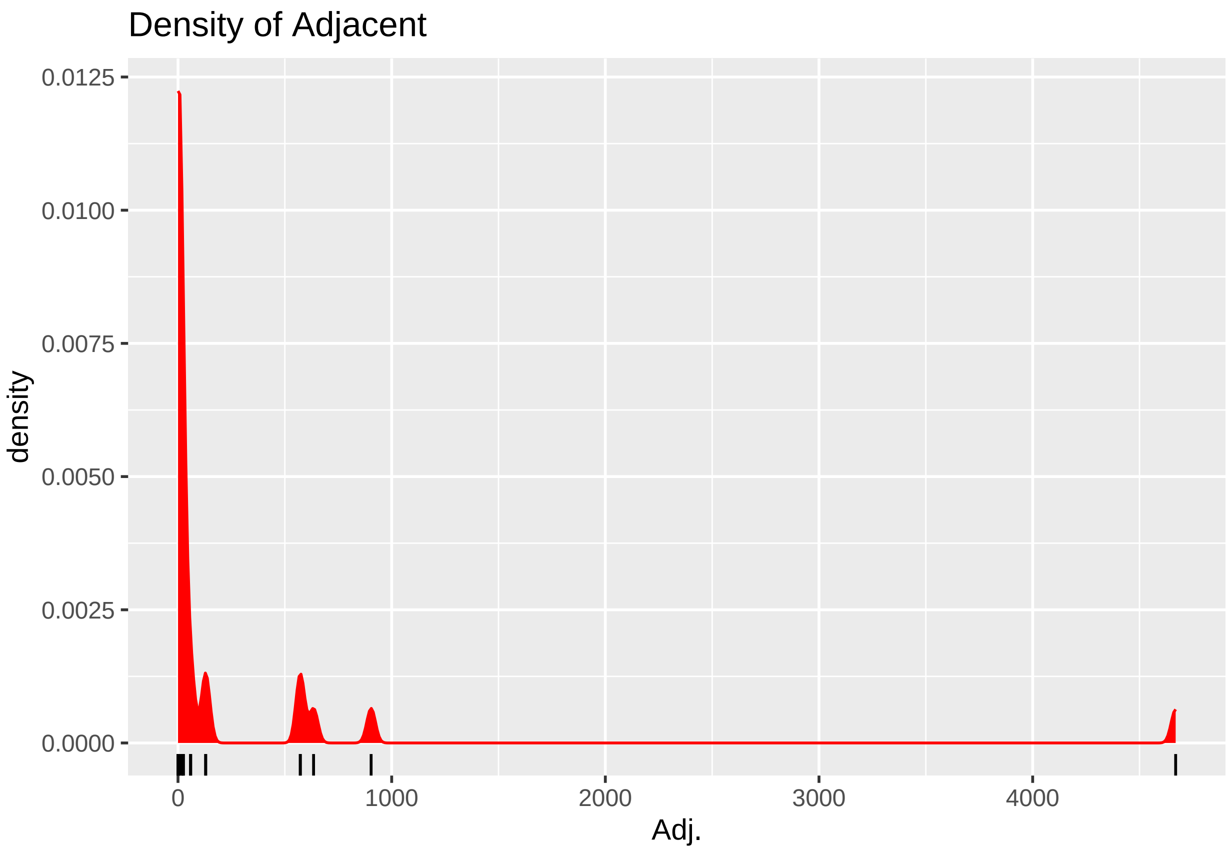

```r
ggplot(gala, aes(x = Scruz)) +
  geom_density(color = "red", fill = "red") +
  geom_rug() +
  labs(title = "Density of Scruz", x = "scruz")
```




```r
# Adjacent
ggplot(gala, aes(x = Adjacent)) +
  geom_histogram(binwidth = 100, color = "black", fill = "blue") +
  labs(title = "Dist. Adjacent", x = "adjacent", y = "freq")
```




```r
ggplot(gala, aes(x = Adjacent)) +
  geom_density(color = "red", fill = "red") +
  geom_rug() +
  labs(title = "Density of Adjacent", x = "Adj.")
```



Our plots and our call of summary(gala) reveals the following information:

```
- (Endemics): Our response variable, Endemics, shows high variability with a large range (0 to 95). Thi

- (Area, Adjacent): Both variables show significant variation, with a high max compared to mean values,

- (Scruz): Also shows moderately high variation, outliers can be seen in distribution graph however the

- (Elevation, Nearest): Both show more balanced distributions compared to our other variables, but stil
```

---

2. Is at least one of the predictors useful in predicting the response (Endemics)?

Establishing an alpha (α) value of 0.05.

Code:

```
# getting p-vals
model <- lm(Endemics ~ Area + Elevation + Nearest + Scruz + Adjacent, data = gala)
summary(model)
```

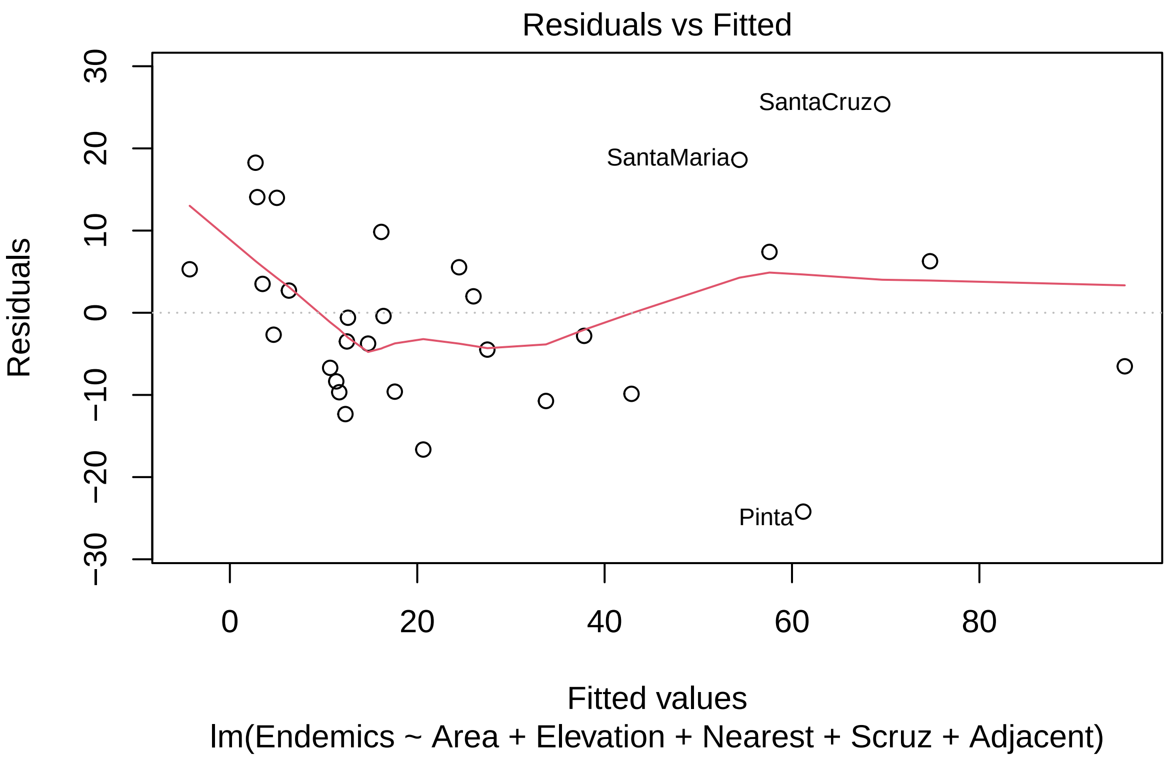
```
##
## Call:
## lm(formula = Endemics ~ Area + Elevation + Nearest + Scruz +
##     Adjacent, data = gala)
```

```
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -24.201  -7.941  -1.637   6.086  25.376
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.099665   3.859507   1.321   0.1989
## Area        -0.008466   0.004518  -1.874   0.0732 .
## Elevation    0.083530   0.010813   7.725 5.83e-08 ***
## Nearest      0.025173   0.212405   0.119   0.9066
## Scruz       -0.056623   0.043403  -1.305   0.2044
## Adjacent    -0.017438   0.003567  -4.889 5.50e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.29 on 24 degrees of freedom
## Multiple R-squared:  0.8328, Adjusted R-squared:  0.7979
## F-statistic:  23.9 on 5 and 24 DF,  p-value: 1.35e-08
```

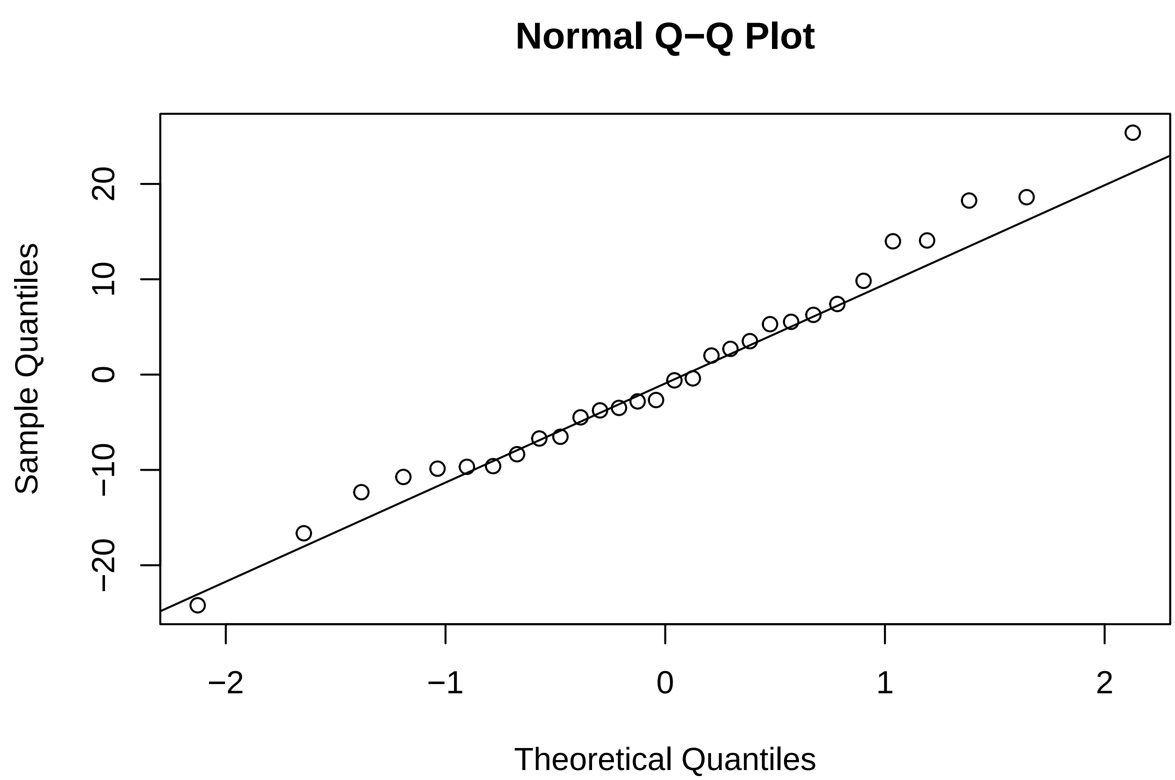
```r
# corelation
cor(gala[, c("Area", "Elevation", "Nearest", "Scruz", "Adjacent")], gala["Endemics"])
```

```
##               Endemics
## Area       0.616979087
## Elevation  0.792904369
## Nearest    0.005994286
## Scruz     -0.154264319
## Adjacent   0.082658026
```

```r
# Checking model assumptions
# residual vs fitted
plot(model, which = 1)
```




```
# qq plot
qqnorm(resid(model))
qqline(residuals(model))
```



Answer:

- Allowing an $\alpha = 0.05$, we can see that…
  - Elevation (p-val: 5.83e-08) and Adjacent (p-val: 5.50e-05) are useful in predicting the response (Endemics). Elevation (0.79) has high Pearson correlation with Endemics signifying a strong linear relationship. However, Adjacent (0.08) has a relatively low Pearson corelation with Endemics, while it still may have a significant relationship with the response, it is unlikely that this relationship is linear. Alternatively Adjacent can be acting as a supressor variable, improving prediction accuracy when combined with other predictors like Elevation.
  - Area (p-val: 0.0732) is slightly above our alpha 0.05, showing that Area has some predictive value but is not as statistically significant as Elevation or Adjacent. Area also shows to have a pearson correlation with our response, suggesting that its relationship with Endemics may be linear.
  - Scruz (p-val: 0.2044) and Nearest (p-val: 0.9066) have little to no meaningful relationship with Endemics that can be leveraged in prediction. Scruz has a negative pearson correlation value of -0.15, which can suggest that islands further from Santa Cruz may have slightly less Endemics, but the value is weak and not statistically meaningful. Nearest has almost 0 pearson correlation with Endemics, and its high p-value means we fail to reject the null hypothesis that Nearest has no effect.

---

3. Perform model selection to determine whether all the predictors help to explain the response.

Code:

```r
# remove pointless vars iteratively based on AIC
stepmodel <- stepAIC(model, direction = "both")
```

```
## Start:  AIC=155.81
## Endemics ~ Area + Elevation + Nearest + Scruz + Adjacent
##
##             Df Sum of Sq     RSS    AIC
## - Nearest    1       2.1  3625.0 153.83
## <none>                    3622.9 155.81
## - Scruz      1     256.9  3879.8 155.87
## - Area       1     530.0  4152.9 157.91
## - Adjacent   1    3608.5  7231.4 174.55
## - Elevation  1    9008.3 12631.2 191.28
##
## Step:  AIC=153.83
## Endemics ~ Area + Elevation + Scruz + Adjacent
##
##             Df Sum of Sq     RSS    AIC
## <none>                    3625.0 153.83
## - Scruz      1     375.6  4000.5 154.79
## + Nearest    1       2.1  3622.9 155.81
## - Area       1     567.3  4192.3 156.19
## - Adjacent   1    3979.1  7604.1 174.06
## - Elevation  1    9595.1 13220.1 190.65
```

```r
summary(stepmodel)
```

```
##
## Call:
## lm(formula = Endemics ~ Area + Elevation + Scruz + Adjacent,
##     data = gala)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -24.083  -8.078  -1.711   5.864  25.212
##
## Coefficients:
##              Estimate Std. Error t value Pr(>|t|)
## (Intercept)  5.119366   3.779125   1.355   0.1876
## Area        -0.008575   0.004335  -1.978   0.0591 .
## Elevation    0.083829   0.010305   8.135 1.73e-08 ***
## Scruz       -0.053404   0.033184  -1.609   0.1201
## Adjacent    -0.017558   0.003352  -5.239 2.01e-05 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 12.04 on 25 degrees of freedom
## Multiple R-squared:  0.8327, Adjusted R-squared:  0.8059
## F-statistic:  31.1 on 4 and 25 DF,  p-value: 2.25e-09
```

Answer:

- Using step-wise AIC selection, we can get a final optimal model composed of variables…
  - Adjacent: (p-val: 2.01e-05) shows to be highly significant in explaining the number of endemic species.
  - Elevation: (p-val: 1.73e-08) shows to be highly significant in explaining the number of endemic species.
  - Area: (p-val: 0.0591) is borderline insignificant at our alpha = 0.05 but can still be slightly useful in a less strict model.
  - Scruz: (p-val: 0.1201) is not statistically significant at our alpha = 0.05, and will not help much in predicting the response.
- Our step-wise AIC removed variable Nearest as it was shown to increase AIC, reducing response prediction in our final model.
- In summation, all variables were not useful in predicting the response, and variable Nearest was removed using step-wise AIC, giving a new optimized model with:

  Endemics ~ Area + Elevation + Scruz + Adjacent

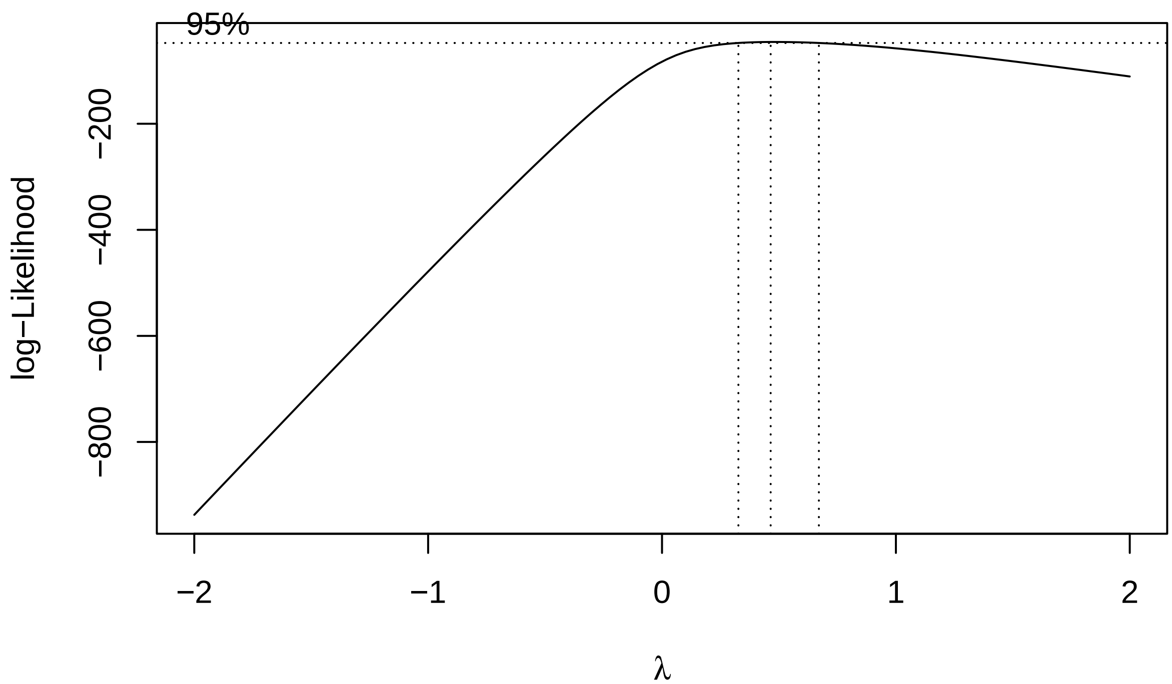
---

4. Perform a Box-Cox transformation to determine if there is a need to transform the response.

Code:

```
# to avoid 0.00value stopping boxcox(), shifting by insignificant amount
gala$Endemics <- gala$Endemics + 0.000001

# boxcox of stepwise AIC optimized model
bc <- boxcox(stepmodel)
```



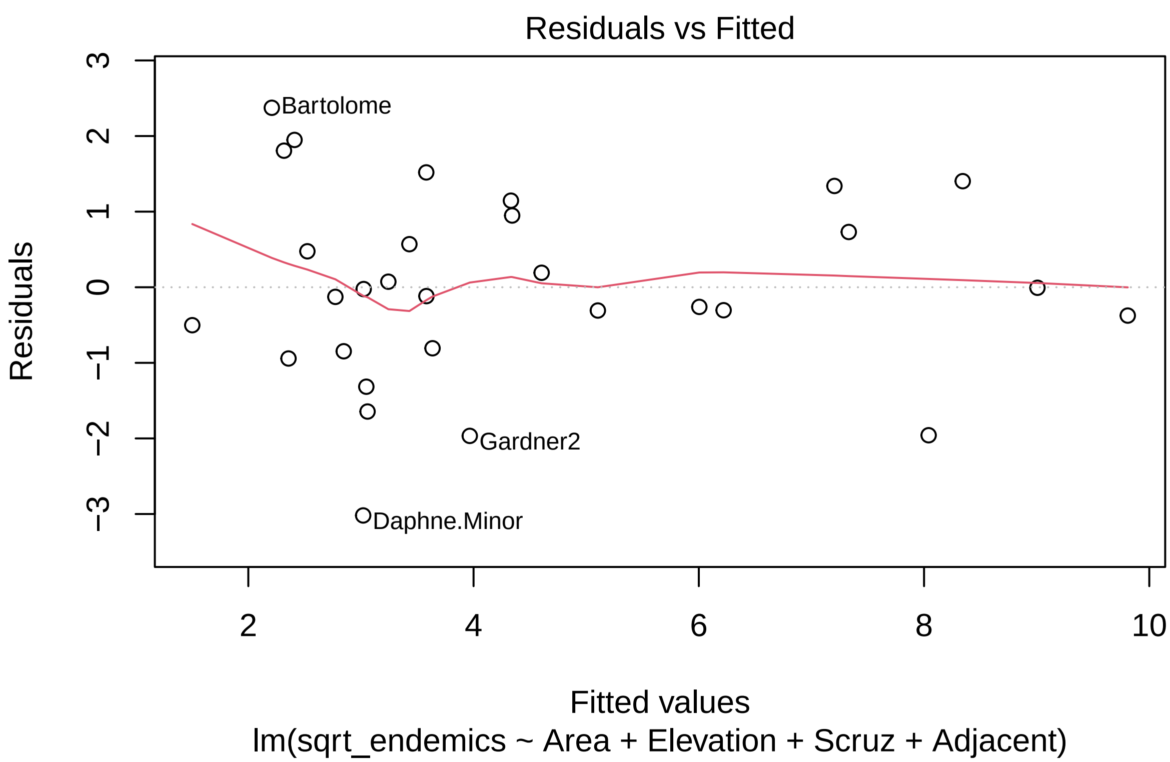
- As we get an approximate λ value of near 0.5, I will apply a square root transformation to our response variable Endemics.

```
sqrt_endemics <- sqrt(gala$Endemics)
bc_sqrt_model <- lm(sqrt_endemics ~ Area + Elevation + Scruz + Adjacent, data = gala)
summary(bc_sqrt_model)
```

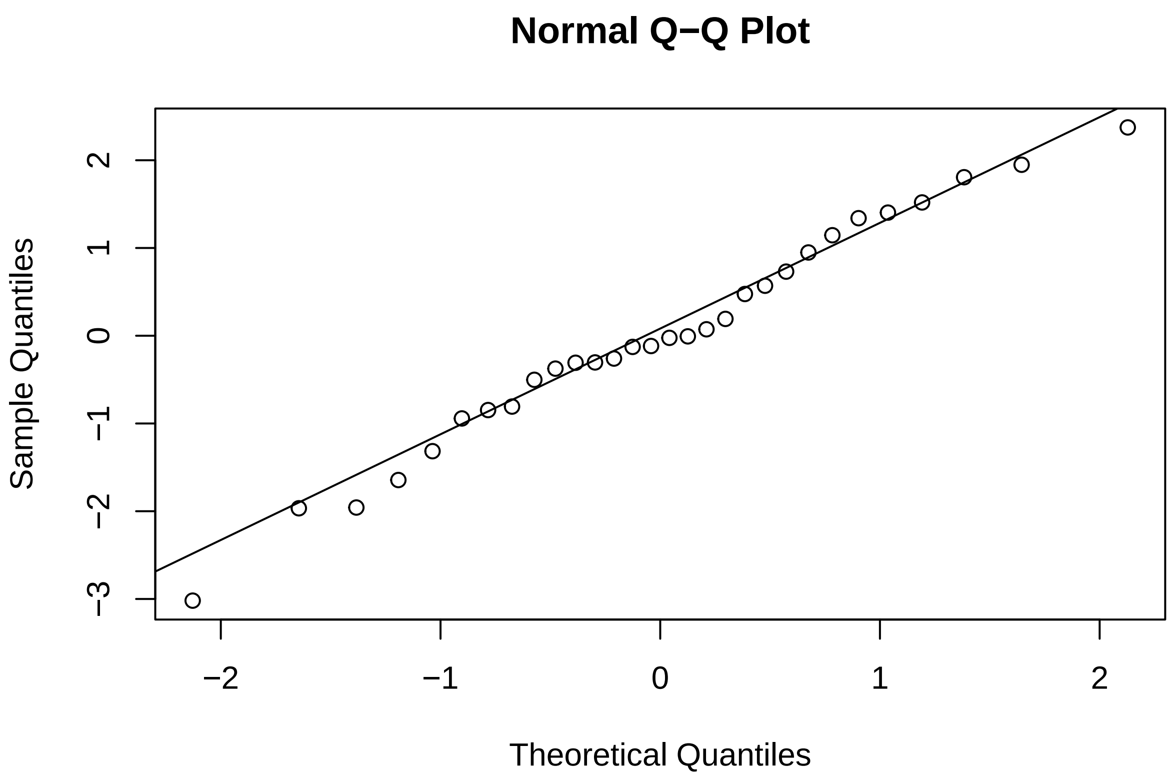
```
##
## Call:
## lm(formula = sqrt_endemics ~ Area + Elevation + Scruz + Adjacent,
##     data = gala)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -3.01882 -0.73102 -0.07077  0.89430  2.37381
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  2.2927254  0.4270514   5.369 1.44e-05 ***
```

```
## Area        -0.0011646  0.0004899  -2.377 0.025415 *
## Elevation    0.0082225  0.0011645   7.061 2.12e-07 ***
## Scruz       -0.0030803  0.0037498  -0.821 0.419154
## Adjacent    -0.0015686  0.0003787  -4.142 0.000344 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1.361 on 25 degrees of freedom
## Multiple R-squared:  0.762,  Adjusted R-squared:  0.7239
## F-statistic: 20.01 on 4 and 25 DF,  p-value: 1.694e-07
```

```r
# residual vs fitted
plot(bc_sqrt_model, which = 1)
```




```r
# normal QQ
qqnorm(resid(bc_sqrt_model))
qqline(residuals(bc_sqrt_model))
```



- Both plots show significant improvements over our original model. There is significantly less spread in Residuals as well as a more linear relationship in our QQ plot.
- Residual plots for both models(original and transformed) showed no significant patterns or heteroscedasticity; transformation doesn't violate assumptions of linear regression.

---

5. Compare a generalized additive model fit with the linear regression fit and explain why you preferred one model over the other.

Code:

```
# setting up GAM model and GAM model without sqrt transformation to Endemics
gam_model <- gam(sqrt_endemics ~ s(Area) + s(Elevation) + s(Scruz) + s(Adjacent), data = gala)
gam_model_no_sqrt <- gam(Endemics ~ s(Area) + s(Elevation) + s(Scruz) + s(Adjacent), data = gala)

summary(gam_model)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## sqrt_endemics ~ s(Area) + s(Elevation) + s(Scruz) + s(Adjacent)
##
## Parametric coefficients:
```

```
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   4.4290     0.1894   23.39 2.17e-14 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                 edf Ref.df      F  p-value
## s(Area)       1.000  1.000  0.000    0.999
## s(Elevation) 2.535  3.203 29.044 1.04e-08 ***
## s(Scruz)     7.423  8.127  2.009    0.107
## s(Adjacent)  1.000  1.000  1.968    0.179
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =   0.84   Deviance explained = 90.6%
## GCV = 1.8941  Scale est. = 1.076     n = 30
```

```r
summary(gam_model_no_sqrt)
```

```
##
## Family: gaussian
## Link function: identity
##
## Formula:
## Endemics ~ s(Area) + s(Elevation) + s(Scruz) + s(Adjacent)
##
## Parametric coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  26.1000     0.8468   30.82 2.84e-12 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Approximate significance of smooth terms:
##                 edf Ref.df     F  p-value
## s(Area)       2.761  3.091 1.779 0.297968
## s(Elevation) 5.569  6.301 9.714 0.000528 ***
## s(Scruz)     8.355  8.727 5.347 0.005857 **
## s(Adjacent)  1.000  1.000 0.983 0.342691
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## R-sq.(adj) =  0.971   Deviance explained = 98.9%
## GCV = 57.042  Scale est. = 21.513    n = 30
```

- There is a considerable improvement in both of our GAM models:
  - Our GAM model without square root transformaition has a higher Deviance Explained score of 98%, as well as higher adjusted R-squared value of 0.97.
  - Our GAM with square root transformation to response variable Endemics also shows a slightly improved adjusted R-squared value of 0.84 (compared to below 0.80 for all non-GAM models) as well as improved deviance explained.
- This confirms either of the two GAM models will be the better choice moving forward, however, we will need to compare AIC/BIC values to decide which is the best of the two.

```r
# original model vs GAM with sqrt transformation
AIC(model, stepmodel, bc_sqrt_model, gam_model_no_sqrt, gam_model)
```

```
##                         df       AIC
## model              7.00000 242.95108
## stepmodel          6.00000 240.96863
## bc_sqrt_model      6.00000 110.14803
## gam_model_no_sqrt 19.68558 187.31417
## gam_model         13.95834  98.28344
```

```r
BIC(model, stepmodel, bc_sqrt_model, gam_model_no_sqrt, gam_model)
```

```
##                         df      BIC
## model              7.00000 252.7595
## stepmodel          6.00000 249.3758
## bc_sqrt_model      6.00000 118.5552
## gam_model_no_sqrt 19.68558 214.8975
## gam_model         13.95834 117.8418
```

- GAM without sqrt(Endemics) outperforms in terms of adjusted R-squared values and Deviance Explained.
- GAM model with sqrt(Endemics) significantly outperforms GAM (no sqrt) in terms of AIC and BIC.
  - This indicating it is less likely to overfit.
  - Lower AIC/BIC values indicates better balance between fit and complexity.
- GAM with transformation of sqrt(Endemics) may be the safer and more generalizable model for making predictions on Endemics.
- While the GAM model without transformation scores higher on both adjusted R-squared and deviance explained, its complexity (df = 19) and much higher AIC/BIC criterion values suggests that it may not perform well on new data.
- Thus, the transformed GAM provides more robust and justifiable model for our conclusion.

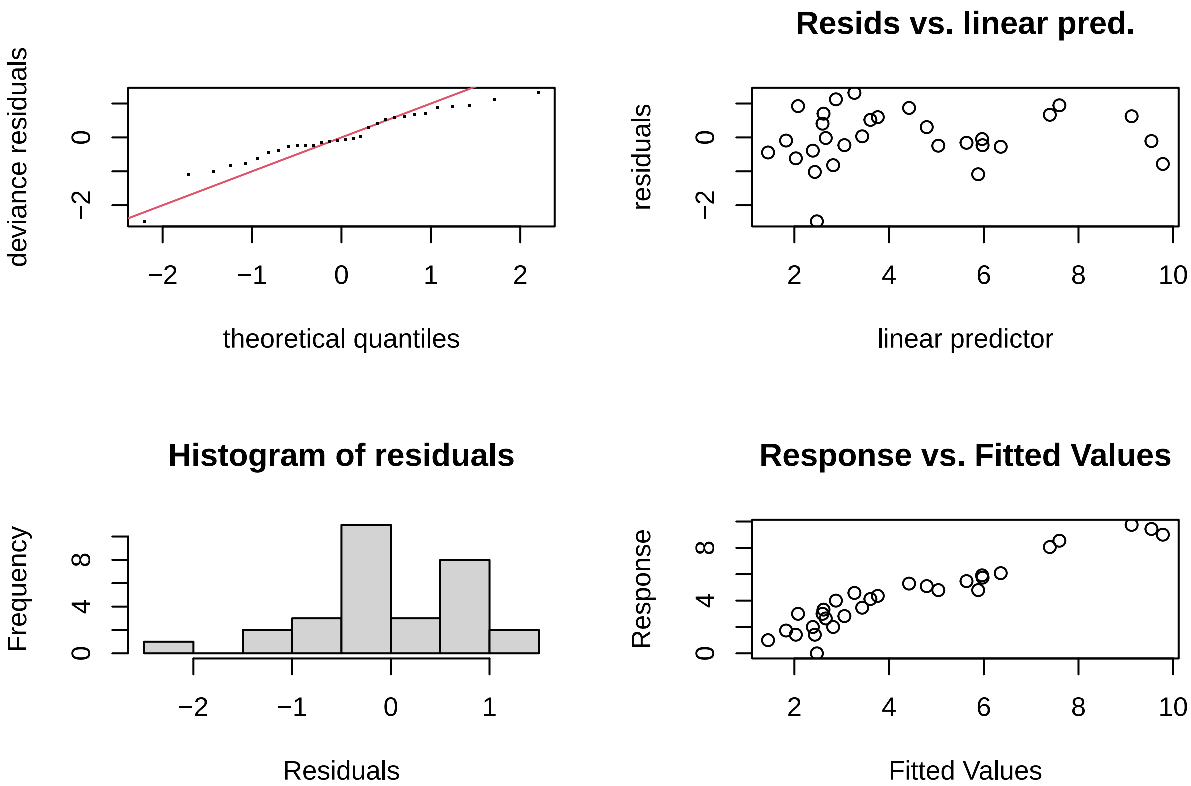
---

6. How well does the chosen model fit the data? Quantify the fit numerically. Additionally, check the model assumptions and summarize your findings.

- Untransformed GAM has MUCH better scores for R-squared and Deviance Explanined, but it is marginal compared to the scores for the transformed GAM.
- GAM with sqrt transformation has much lower AIC/BIC values (98.3, 117.8) showing it is better at fitting the data.
  - Has significantly lower AIC/BIC criterion values, indication a much more flexible model.
  - Lower degrees of freedom shows reduced risk of overfitting.
  - Trade-off in adjusted R-squared and deviance explained is acceptable as transformed GAM is much better at generalizing the trends.

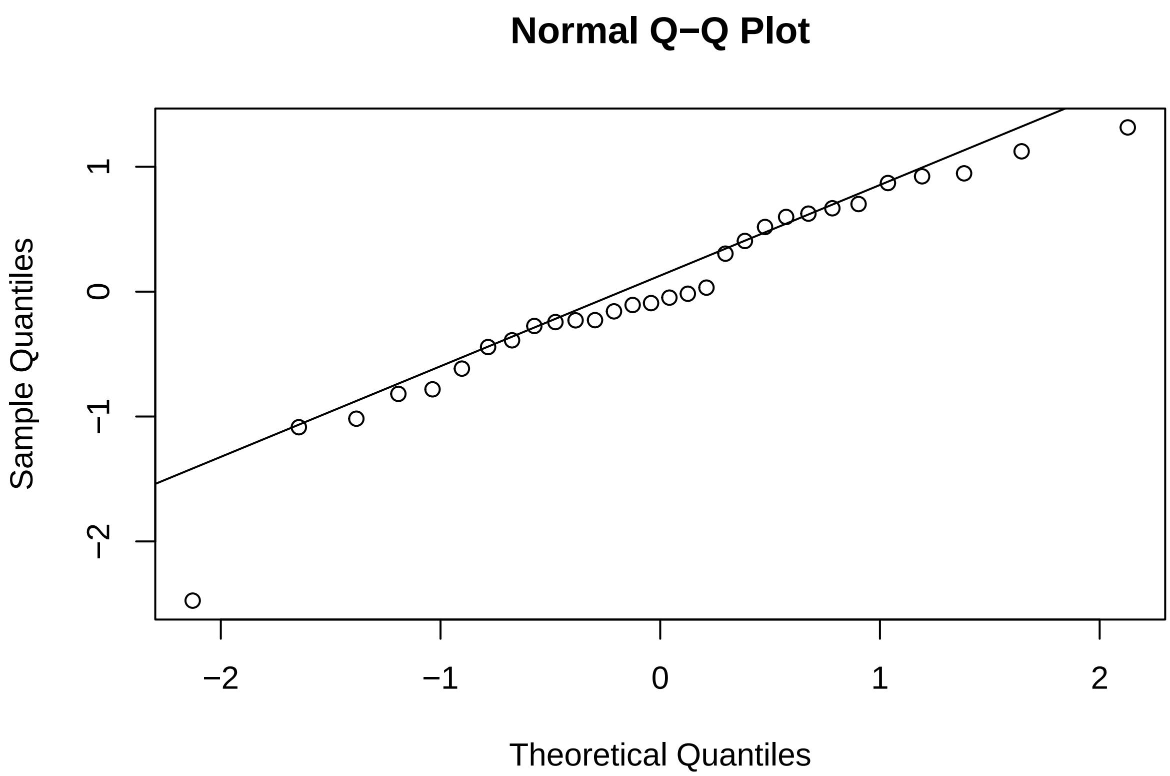
Code:

```r
# residual vs fitted
gam.check(gam_model)
```




```
##
## Method: GCV   Optimizer: magic
## Smoothing parameter selection converged after 15 iterations.
## The RMS GCV score gradient at convergence was 1.918097e-07 .
## The Hessian was positive definite.
## Model rank =  37 / 37
##
## Basis dimension (k) checking results. Low p-value (k-index<1) may
## indicate that k is too low, especially if edf is close to k'.
##
##                 k'  edf k-index p-value
## s(Area)       9.00 1.00    1.28    0.88
## s(Elevation) 9.00 2.53    1.14    0.71
## s(Scruz)      9.00 7.42    1.25    0.90
## s(Adjacent)  9.00 1.00    0.97    0.38
```

```r
# normal QQ
qqnorm(residuals(gam_model))
qqline(residuals(gam_model))
```



- Residual plots for GAM (no transformation to Endemics) showed no significant patterns or heteroscedasticity; model doesn't violate assumptions of linear regression.
- Our QQ plots show our residuals to be normally distributed, showing no violation in model assumptions.

---

7. Predict the response value when all the chosen predictors take their median value. Quantify the prediction uncertainty.

Code:

```
median_values <- data.frame(
  Area = median(gala$Area, na.rm = TRUE),
  Elevation = median(gala$Elevation, na.rm = TRUE),
  Scruz = median(gala$Scruz, na.rm = TRUE),
  Adjacent = median(gala$Adjacent, na.rm = TRUE)
)

prediction <- predict(
  gam_model_no_sqrt,
  newdata = median_values,
  se.fit = TRUE
)

fit <- prediction$fit
se <- prediction$se.fit
```

```r
alpha <- 0.05
z_value <- qnorm(1 - alpha / 2)
lower_ci <- fit - z_value * se
upper_ci <- fit + z_value * se

pred_uncertainty <- (upper_ci - lower_ci) / 2

cat("Predicted Endemics:", round(fit, 2), "\n")
```

```
## Predicted Endemics: 17.47
```

```r
cat("95% Prediction Interval: [", round(lower_ci, 2), ",", round(upper_ci, 2), "]\n")
```

```
## 95% Prediction Interval: [ 10.53 , 24.4 ]
```

```r
cat("Prediction Uncertainty: ", pred_uncertainty)
```

```
## Prediction Uncertainty:  6.936156
```

- Prediction for Endemics for basic GAM model is 17.47, with a 95% Prediction Interval of [10.53, 24.4], and a Prediction uncertainty of 6.936.

```r
median_values <- data.frame(
  Area = median(gala$Area, na.rm = TRUE),
  Elevation = median(gala$Elevation, na.rm = TRUE),
  Scruz = median(gala$Scruz, na.rm = TRUE),
  Adjacent = median(gala$Adjacent, na.rm = TRUE)
)

prediction <- predict(
  gam_model,
  newdata = median_values,
  se.fit = TRUE
)

fit_sqrt <- prediction$fit
se_sqrt <- prediction$se.fit

alpha <- 0.05
z_value <- qnorm(1 - alpha/2)
lower_ci_sqrt <- fit_sqrt - z_value * se_sqrt
upper_ci_sqrt <- fit_sqrt + z_value * se_sqrt

# back-transform (square)
fit_endemics <- fit_sqrt^2
lower_ci_endemics <- lower_ci_sqrt^2
upper_ci_endemics <- upper_ci_sqrt^2

pred_uncertainty <- (upper_ci_endemics - lower_ci_endemics) / 2

cat("Predicted Endemics:", round(fit_endemics, 2), "\n")
```

```
## Predicted Endemics: 12.66
```

```r
cat("95% CI for Endemics: [", round(lower_ci_endemics, 2), ",", round(upper_ci_endemics, 2), "]\n")
```

```
## 95% CI for Endemics: [ 6.32 , 21.17 ]
```

```r
cat("Prediction Uncertainty: ", pred_uncertainty)
```

```
## Prediction Uncertainty:  7.424341
```

- Prediction for Endemics for transformed GAM model is 12.66 with a 95% Confidence Interval of [6.32, 21.17], and prediction uncertainty is 7.424.
- While basic GAM shows a lower prediction uncertainty, our transformed GAM is still preferred due to its generalizability to new data. Similarly to that of adjusted R-square and deviance explained, the trade-off in prediction uncertain is minimal and acceptable.

---

8. Provide a thoughtful conclusion that summarize your findings.

Answer:

- In this analysis of species diversity on the Galapagos Islands, I explored how geographic and topographic factors influence the number of Endemic species. Descriptive summaries of the raw data revealed substantial variability and right-skewed distributions, particularly in variables 'Area' and 'Adjacent', suggesting the presence of outliers.
- The regression analysis shows that 'Elevation' and 'Adjacent' are significant predictors of our response variable 'Endemics', with 'Elevation' having the strongest association. 'Area' was close to significance and warrants further consideration, but 'Scruz' and 'Nearest' are not particularly informative. Stepwise AIC selection confirmed 'Nearest' to be removed as it slightly improved model performance.
- Box-Cox transformation using a square-root transformation on the response variable improves normality of residuals and reduces heterogeneity of variance. Comparing linear regression to generalized additive models (GAM), we found that the GAM without transformation yielded the highest adjusted R^2 (0.97) and deviance explained (98.9%), indicating a strong fit. However, this came with a risk of over-fitting due to increased complexity. The GAM with a transformed response offered a better balance between model complexity and generalizability, as reflected by lower AIC/BIC values and a more stable prediction uncertainty.
- Ultimately, our transformed GAM model was chosen as the best choice for making future predictions on 'Endemics'.

Problem 2: Salaries

- Perform a detailed analysis using the Salaries data set to comment on the salary differences between male and female faculty members.

Code:

```r
library(carData)
data(Salaries)
```

Data Format

References: Fox J. and Weisberg, S. (2019) An R Companion to Applied Regression, Third Edition, Sage.

A data frame with 397 observations on the following 6 variables.

rank - a factor with levels 'AssocProf', 'AsstProf', 'Prof'.

discipline - a factor with levels A ("theoretical" departments) or B ("applied" departments).

yrs.since.phd - years since PhD.

yrs.service - years of service.

sex - a factor with levels Female Male

salary - nine-month salary, in dollars.

---

1. Provide a descriptive summary of the data set.

Code:

```r
head(Salaries)
```

```
##        rank discipline yrs.since.phd yrs.service  sex salary
## 1      Prof          B            19          18 Male 139750
## 2      Prof          B            20          16 Male 173200
## 3  AsstProf          B             4           3 Male  79750
## 4      Prof          B            45          39 Male 115000
## 5      Prof          B            40          41 Male 141500
## 6 AssocProf          B             6           6 Male  97000
```

```r
summary(Salaries)
```

```
##         rank     discipline yrs.since.phd    yrs.service        sex
##  AsstProf : 67   A:181      Min.   : 1.00   Min.   : 0.00   Female: 39
##  AssocProf: 64   B:216      1st Qu.:12.00   1st Qu.: 7.00   Male  :358
##  Prof     :266              Median :21.00   Median :16.00
##                             Mean   :22.31   Mean   :17.61
##                             3rd Qu.:32.00   3rd Qu.:27.00
##                             Max.   :56.00   Max.   :60.00
##      salary
##  Min.   : 57800
##  1st Qu.: 91000
##  Median :107300
##  Mean   :113706
##  3rd Qu.:134185
##  Max.   :231545
```

Answer:

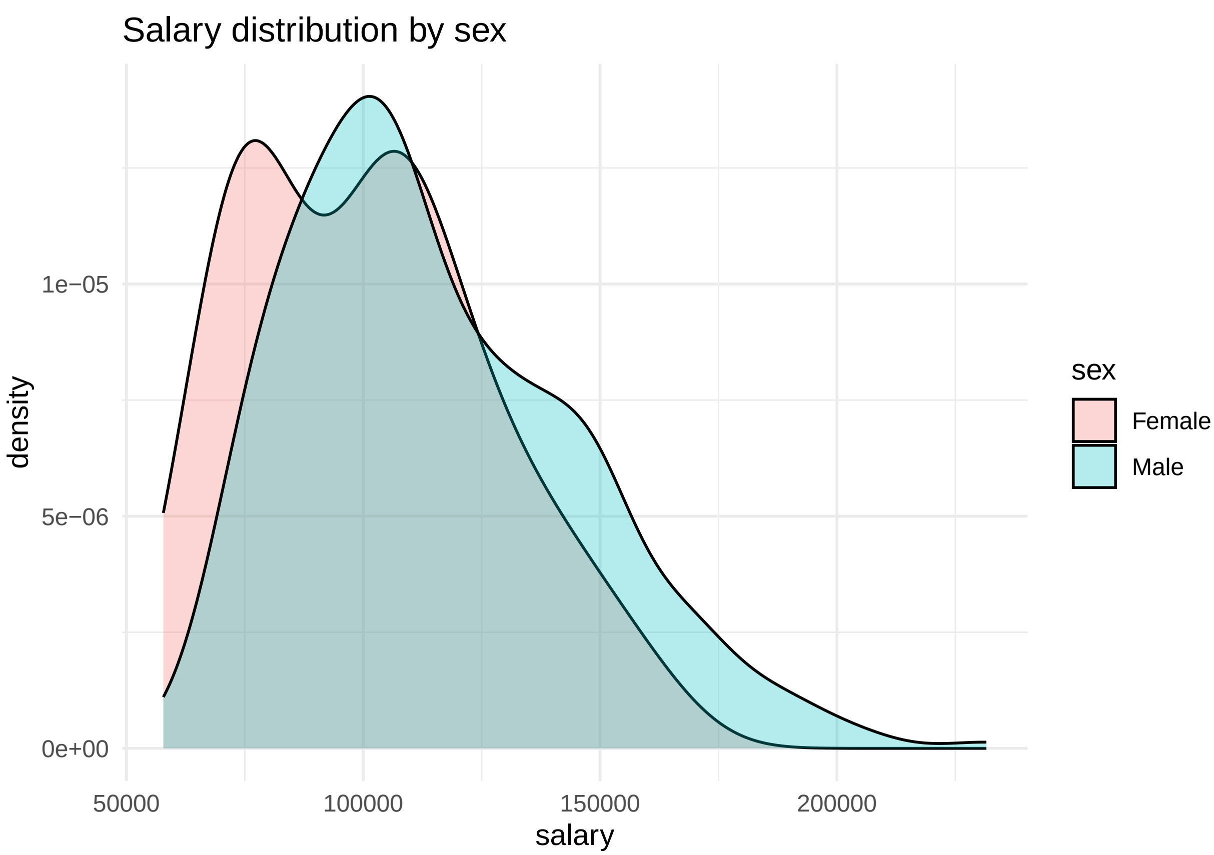
- 'salary' (Response variable)
  - Salary ranges from 57k to 231k, with a median salary of 107k and a slightly higher mean salary of 113k, indicating a right skewed distribution.
  - This spread is likely linked to faculties 'rank', 'discipline', 'yrs.since.phd', 'yrs.service', and possibly even 'sex'.
- 'rank'
  - Most faculty members are Professors (266).
  - Associate Professors (64) and Assistant Professors (67) combined make up slightly less than half of the total faculty.
- 'discipline'
  - There are two disciplines, A (Theoretical) and B (Appied), with Applied discipline (216) being slightly more popular than Theoretical discipline (181).
- 'yrs.since.phd'
  - Staff have an average of 22.31 years since they completed their PhD, with a close median value of 21 years.
  - There is a large range between the min and max values, indicating that the staff has a wide range of experience, with some staff haing 56 years since completeing their PhD while others have only 1 year since PhD.
- 'yrs.service'
  - Staff has an average of 17.6 years of service with their institution, with a median of 16 years.
  - Simmilar to before, there is a wide range of experience, with some being new to the institution (0.00 yrs.service) and others spending up to 60 years at the institution.
- 'sex'
  - There is an immense gender imbalance, Male faculty make up roughly 90% of the staff (358), whereas Female faculty make up only slightly less than 10% (39) of the staff roster.
  - The discrepancy could potentially impact the reliability of statistical analyses regarding salary variations, and it is essential to take note of this factor during interpretation.
- Initial Observations:
  - Male dominance in dataset (~90%) may bias salary averages if male faculty disproportionatly holds higher 'rank', 'yrs.since.phd', or 'yrs.service'.
  - It is probable that 'rank', 'yrs.since.phd' and 'yrs.service' could serve as confounding elements. The observed salary differences between genders may be attributed to discrepancies in these aspects to some degree.
  - Right skew observed in salary distribution indicates presence of high salary outliers, and can even potentially affect comparison between male and female salaries, especially because Females make up such a small amount of the data.

---

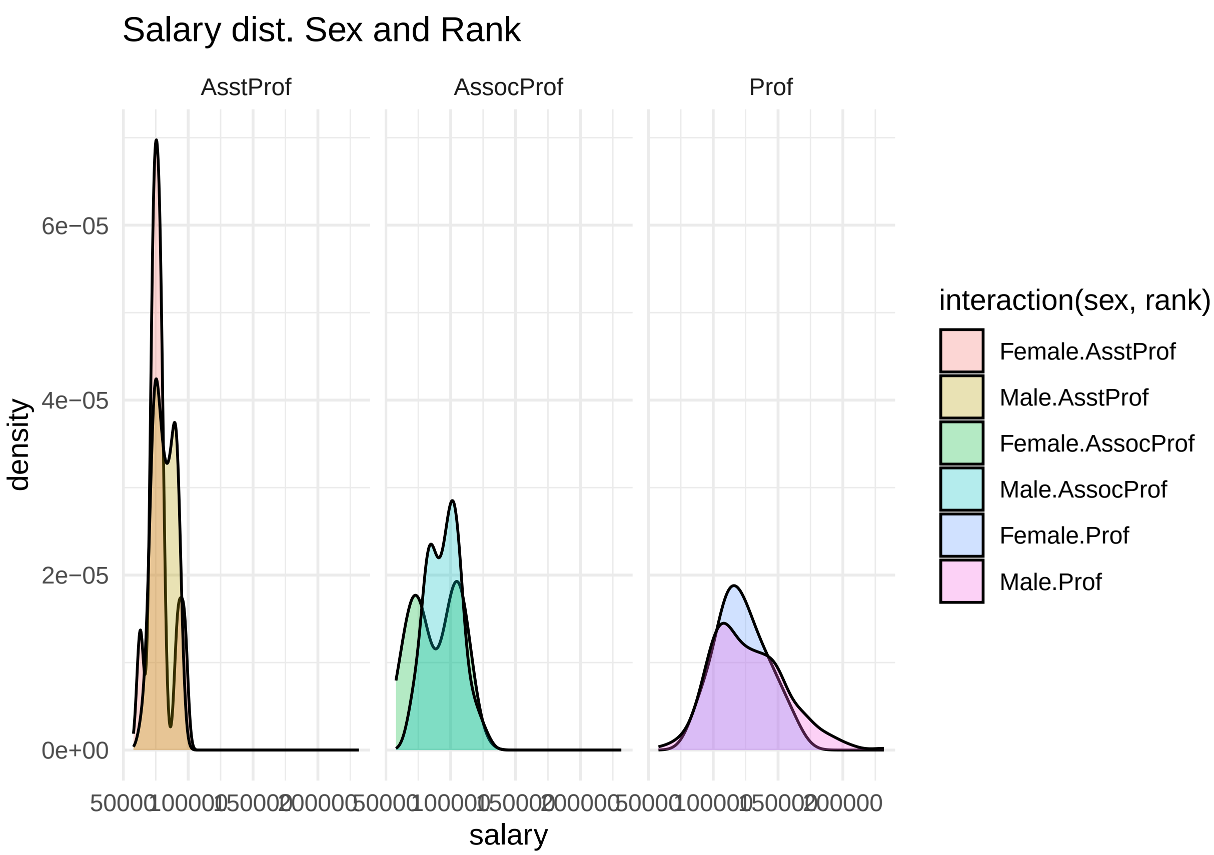
2. Comment on whether there is a need to include interactions between these categorical predictors. Select your model and justify your choice.

Code:

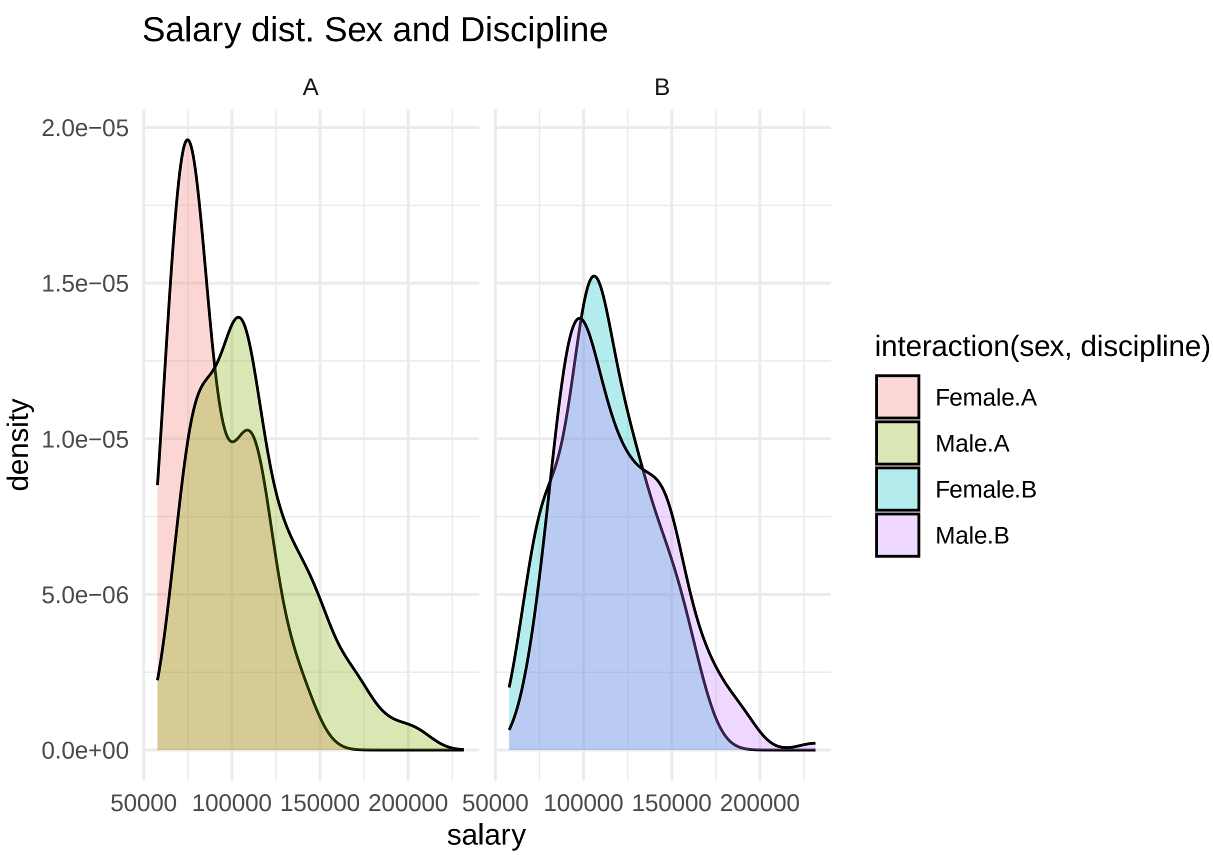
```
# salary dist by sex
ggplot(Salaries, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.3) +
  labs(title = "Salary distribution by sex", x = "salary", y = "density") +
  theme_minimal()
```



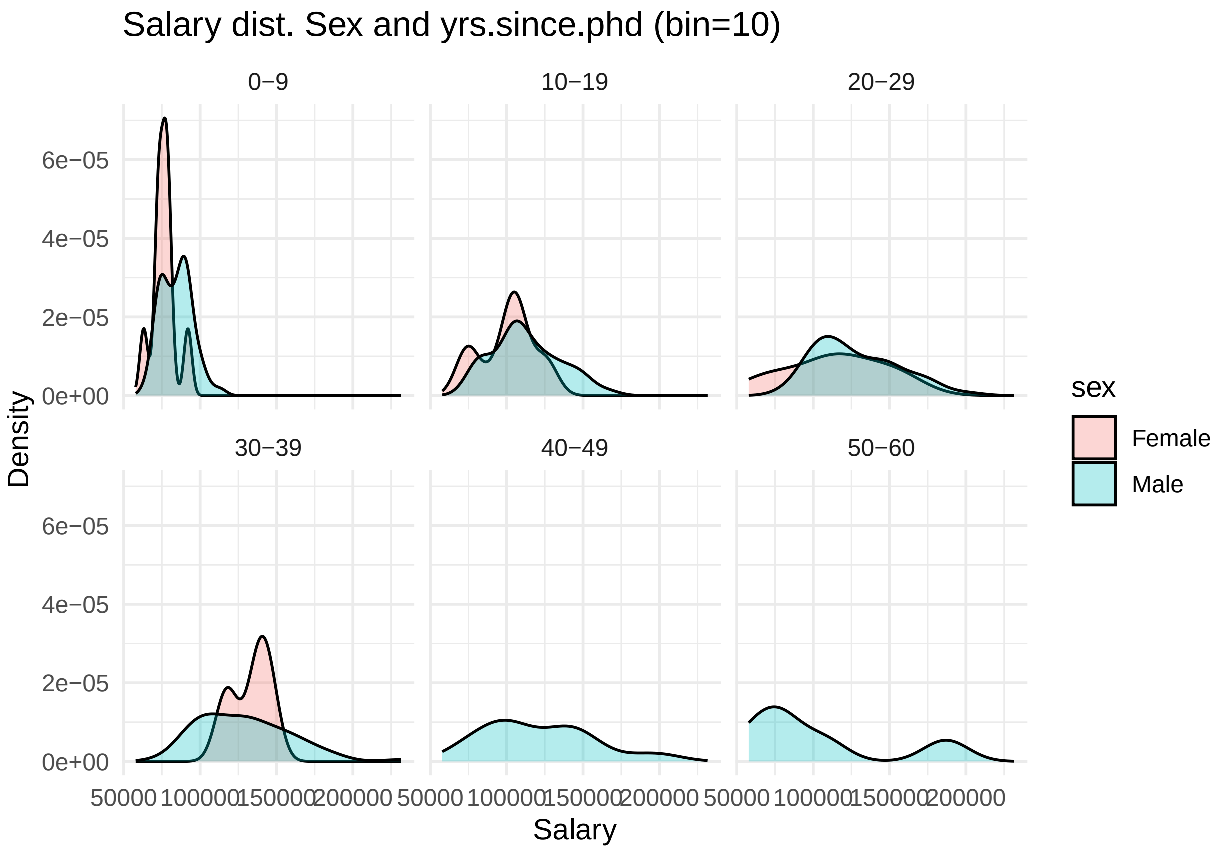

```
# sex and rank
ggplot(Salaries, aes(x = salary, fill = interaction(sex, rank))) +
  geom_density(alpha = 0.3) +
  facet_wrap(~rank) +
  labs(title = "Salary dist. Sex and Rank", x = "salary", y = "density") +
  theme_minimal()
```




```r
# sex and discipline
ggplot(Salaries, aes(x = salary, fill = interaction(sex, discipline))) +
  geom_density(alpha = 0.3) +
  facet_wrap(~discipline) +
  labs(title = "Salary dist. Sex and Discipline", x = "salary", y = "density") +
  theme_minimal()
```



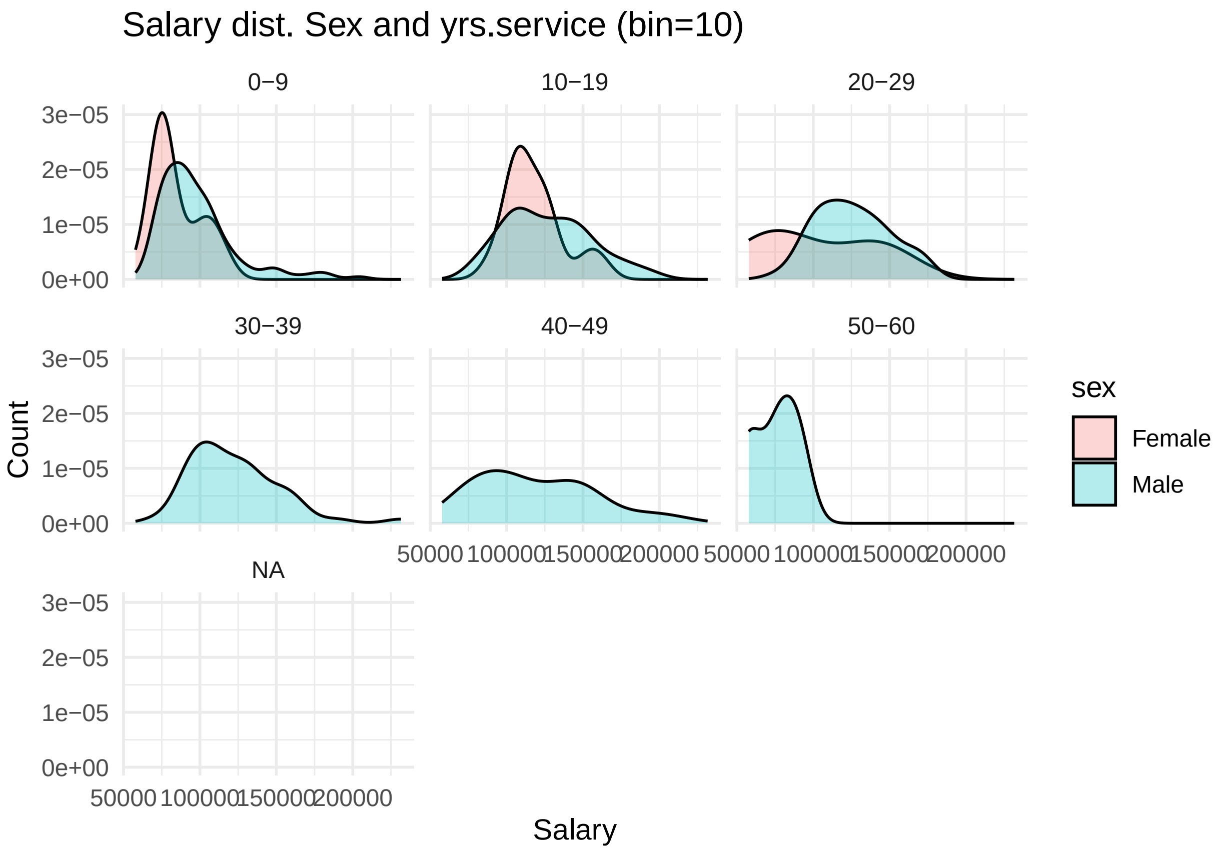

```r
# salary dist: sex and yrs.since.phd
ggplot(Salaries, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.3) +
  facet_wrap(~ cut(yrs.since.phd, breaks = seq(0, 60, by = 10), right = FALSE,
                   labels = c("0-9", "10-19", "20-29", "30-39", "40-49", "50-60"))) +
  labs(title = "Salary dist. Sex and yrs.since.phd (bin=10)", x = "Salary", y = "Density") +
  theme_minimal()
```




```r
# salary dist: sex, yrs.service
ggplot(Salaries, aes(x = salary, fill = sex)) +
  geom_density(alpha = 0.3) +
  facet_wrap(~ cut(yrs.service,
                   breaks = seq(0, 60, by = 10),
                   right = FALSE,
                   labels = c("0-9", "10-19", "20-29", "30-39", "40-49", "50-60")), drop = FALSE) +
  labs(title = "Salary dist. Sex and yrs.service (bin=10)", x = "Salary", y = "Count") +
  theme_minimal()
```

```
## Warning: Groups with fewer than two data points have been dropped.
## Groups with fewer than two data points have been dropped.
```

```
## Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning
## -Inf
## Warning in max(ids, na.rm = TRUE): no non-missing arguments to max; returning
## -Inf
```




```r
# contingency table for pairwise
sex_rank <- table(Salaries$sex, Salaries$rank)
sex_disc <- table(Salaries$sex, Salaries$discipline)
rank_disc <- table(Salaries$rank, Salaries$discipline)

sex_rank
```

```
##
##          AsstProf AssocProf Prof
##   Female       11        10   18
##   Male         56        54  248
```

```r
sex_disc
```

```
##
##            A   B
##   Female  18  21
##   Male   163 195
```

```r
rank_disc
```

```
##
##               A   B
##   AsstProf   24  43
##   AssocProf  26  38
##   Prof      131 135
```

```r
# chi test for association
chisq.test(sex_rank)
```

```
##
##  Pearson's Chi-squared test
##
## data:  sex_rank
## X-squared = 8.5259, df = 2, p-value = 0.01408
```

```r
chisq.test(sex_disc)
```

```
##
##  Pearson's Chi-squared test with Yates' continuity correction
##
## data:  sex_disc
## X-squared = 2.0875e-29, df = 1, p-value = 1
```

```r
chisq.test(rank_disc)
```

```
##
##  Pearson's Chi-squared test
##
## data:  rank_disc
## X-squared = 4.6487, df = 2, p-value = 0.09785
```

- (Sex, Rank):
  - In the event that salary disparities between Male and Female are more pronounced at higher ranks (like full professors compared to assistant professors) it becomes necessary to explore an interaction between 'sex' and 'rank'. Additionally, our Pearson Chi-squared test returns a p-value of $0.01408 < \text{alpha} = 0.05$, confirming significant association between sex and rank.
- (Sex, Discipline):
  - Our Pearson Chi-squared for this pairwise returns a p-value of 1, significantly greater than our alpha. However, in the event that 'Male' and 'Female' staff face salary discrepancies among fields of study, with more significant gaps apparent in 'applied' disciplines versus 'theoretical', it becomes essential to consider an interaction between discipline and sex.
- To account for all confounding variables like 'rank', 'discipline', and 'yrs.since.phd'/'yrs.service', I will use a basic linear regression model:

  Salary ~ sex + rank + discipline + yrs.since.phd + yrs.service

- I will also establish a secondary model to test for overall salary differences for sex with focus on interaction between confounding variables 'rank' and 'discipline'.

  Salary ~ sex * rank + sex * discipline + yrs.since.phd + yrs.service

```r
model_basic <- lm(salary ~ sex + rank + discipline + yrs.since.phd + yrs.service, data=Salaries)
model_rank_disc <- lm(salary ~ sex * rank + sex * discipline + yrs.since.phd + yrs.service, data=Salari

summary(model_basic)
```

```
##
## Call:
## lm(formula = salary ~ sex + rank + discipline + yrs.since.phd +
##     yrs.service, data = Salaries)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -65248 -13211  -1775  10384  99592
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    65955.2     4588.6  14.374  < 2e-16 ***
## sexMale         4783.5     3858.7   1.240  0.21584
## rankAssocProf  12907.6     4145.3   3.114  0.00198 **
## rankProf       45066.0     4237.5  10.635  < 2e-16 ***
## disciplineB    14417.6     2342.9   6.154 1.88e-09 ***
## yrs.since.phd    535.1      241.0   2.220  0.02698 *
## yrs.service     -489.5      211.9  -2.310  0.02143 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22540 on 390 degrees of freedom
## Multiple R-squared:  0.4547, Adjusted R-squared:  0.4463
## F-statistic:  54.2 on 6 and 390 DF,  p-value: < 2.2e-16
```

```r
summary(model_rank_disc)
```

```
##
## Call:
## lm(formula = salary ~ sex * rank + sex * discipline + yrs.since.phd +
##     yrs.service, data = Salaries)
##
## Residuals:
##    Min     1Q Median     3Q    Max
## -64874 -13468  -1382   9888  99935
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            66450.1     7670.5   8.663  < 2e-16 ***
## sexMale                 4002.5     8317.9   0.481  0.63065
## rankAssocProf           6466.1    10010.8   0.646  0.51872
## rankProf               39243.2     9006.7   4.357 1.69e-05 ***
## disciplineB            21616.0     7323.3   2.952  0.00335 **
## yrs.since.phd            540.9      241.6   2.239  0.02572 *
## yrs.service             -500.5      212.6  -2.355  0.01903 *
## sexMale:rankAssocProf   7479.0    10820.1   0.691  0.48985
## sexMale:rankProf        6639.4     9326.3   0.712  0.47695
```

```
## sexMale:disciplineB    -7873.2     7693.9  -1.023  0.30680
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 22580 on 387 degrees of freedom
## Multiple R-squared:  0.4569, Adjusted R-squared:  0.4442
## F-statistic: 36.17 on 9 and 387 DF,  p-value: < 2.2e-16
```

```r
# R-squared to test accuracy
print(c("Basic Model adjusted R-squared: ", summary(model_basic)$adj.r.squared))
```

```
## [1] "Basic Model adjusted R-squared: " "0.446287031864428"
```

```r
print(c("Rank/Discipline interaction Model adjusted R-squared: ", summary(model_rank_disc)$adj.r.square
```

```
## [1] "Rank/Discipline interaction Model adjusted R-squared: "
## [2] "0.444226559508176"
```

```r
# AIC to test fit/complexity balance
AIC(model_basic, model_rank_disc)
```

```
##                 df      AIC
## model_basic      8 9093.826
## model_rank_disc 11 9098.235
```

```r
BIC(model_basic, model_rank_disc)
```

```
##                 df      BIC
## model_basic      8 9125.698
## model_rank_disc 11 9142.058
```

- As our basic model shows the best values for adjusted R-squared, and also shows improvement in AIC/BIC criterion values with lower complexity (df=8), it is the better of the two models.

---

3. Perform a log transformation on response to determine if there is any improvement in terms of model fit.

Code:

```r
model_basic_log <- lm(log(salary) ~ sex + rank + discipline + yrs.since.phd + yrs.service, data=Salarie
model_rank_disc_log <- lm(log(salary) ~ sex * rank + sex * discipline + yrs.since.phd + yrs.service, da

# R-squared to test accuracy
print(c("Transformed Basic Model adjusted R-squared: ", summary(model_basic_log)$adj.r.squared))
```

```
## [1] "Transformed Basic Model adjusted R-squared: "
## [2] "0.517497885172332"
```

```r
print(c("Transformed Rank/Discipline interaction Model adjusted R-squared: ", summary(model_rank_disc_lo
```

```
## [1] "Transformed Rank/Discipline interaction Model adjusted R-squared: "
## [2] "0.517081846547093"
```

```r
# AIC to test fit/complexity balance
AIC(model_basic_log, model_rank_disc_log)
```

```
##                      df       AIC
## model_basic_log      8 -222.7779
## model_rank_disc_log 11 -219.5014
```

```r
BIC(model_basic_log, model_rank_disc_log)
```

```
##                      df       BIC
## model_basic_log      8 -190.9064
## model_rank_disc_log 11 -175.6781
```

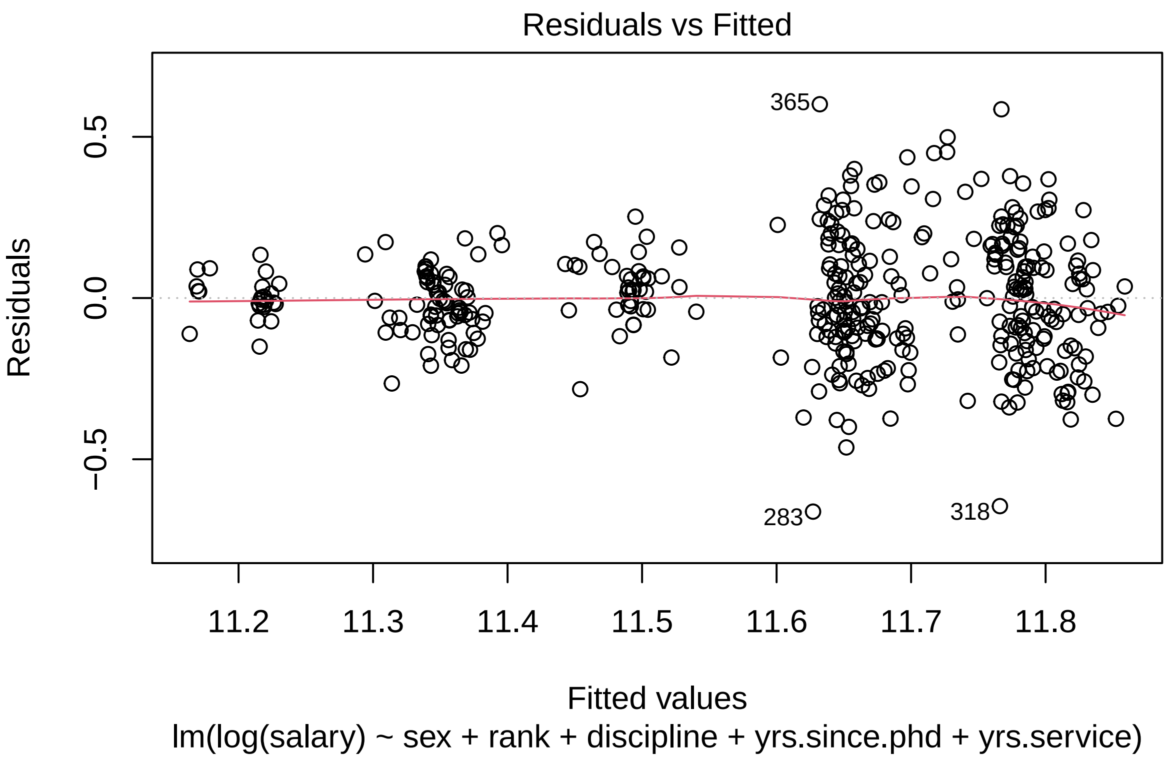
- Log transformation on both models considerably improved model fit, shown in the lower values for Bayesian Information Criterion.
- Due to the extremely similar values for adjusted R-squared and AIC/BIC, we will continue to optimize our basic model instead of focusing on Rank/Discipline interaction for Sex on Salary. This is also in part due to the basic model having less complexity (df=8), thus can be more generalizable.

---

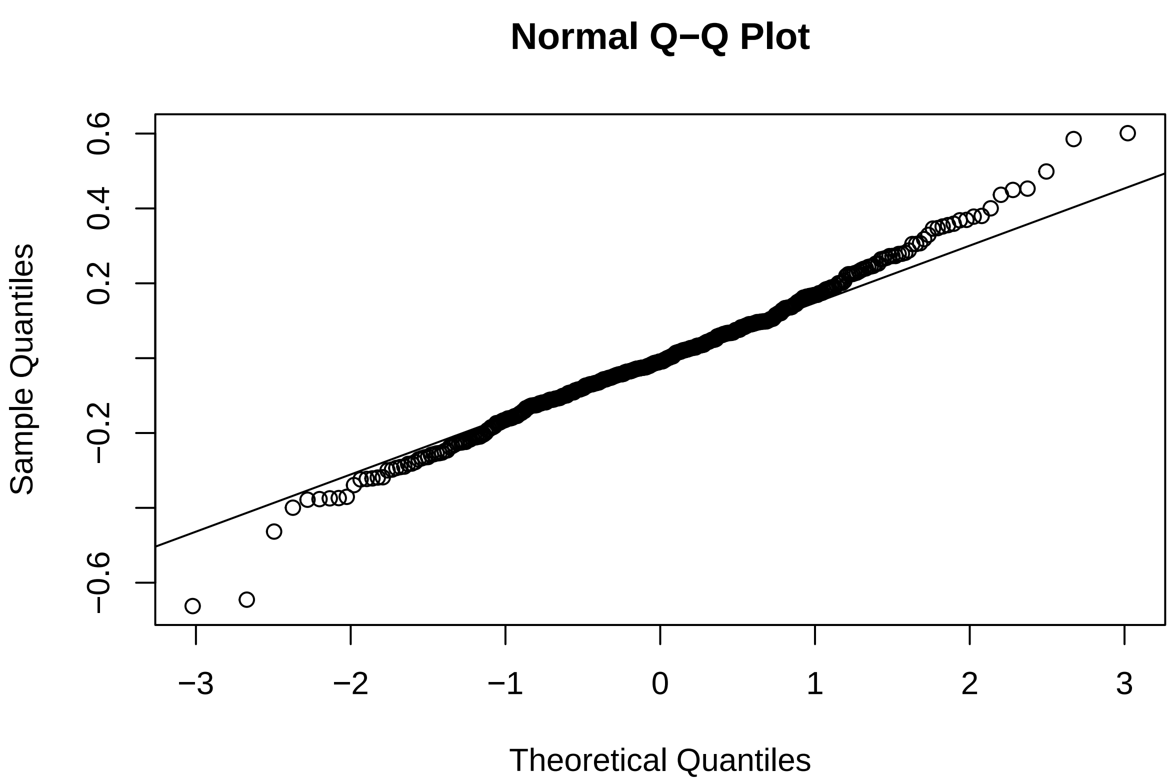
4. How well does the chosen model fit the data? Quantify the fit numerically. Additionally, check the model assumptions and summarize your findings.

- Our model fit improved with the log transformation to response variable 'salary', with significantly improved Akaike and Bayesian Information Criterion values of [A:-222.78, B:-190.91] as opposed to the prior [A:9093.83, B:9125.7].

```r
# Checking model assumptions
# residual vs fitted
plot(model_basic_log, which = 1)
```




```r
# qq plot
qqnorm(resid(model_basic_log))
qqline(residuals(model_basic_log))
```



- Residuals vs. Fitted

  - Residuals appear randomly scattered around 0.0, meets assumption of linearity.
  - Visible clustering reveals variance may not be constant, mildly violating heteroscedasticity assumption.
- QQ Plot
  - Shows residuals to follow a normal distribution, as most points lie near our reference line.
  - Slight deviations on tail indicates mild non-normality of residuals, likely due to outliers.
- Our model meets assumptions of linearity and normality, however there may be some non-constant variance as well as a small amount of outliers.

---

5. Write a thoughtful conclusion to comment on the salary differences between male and female faculty.

- In our analysis, we've revealed a considerable salary disparity between male and female academics across academic ranks. On average, males consistently earn higher salaries compared to their female co-workers, regardless of rank. This trend persists even when considering factors such as discipline.
- Although differences in rank distribution (with more senior positions held by men) contributes partly to the salary disparity, it does not fully explain the observed differences. A deeper investigation, possibly incorporating further variables like 'yrs.since.phd' or 'yrs.service', could show valuable insights into underlying causes.
